

Polar Codes MAC — All Experiments

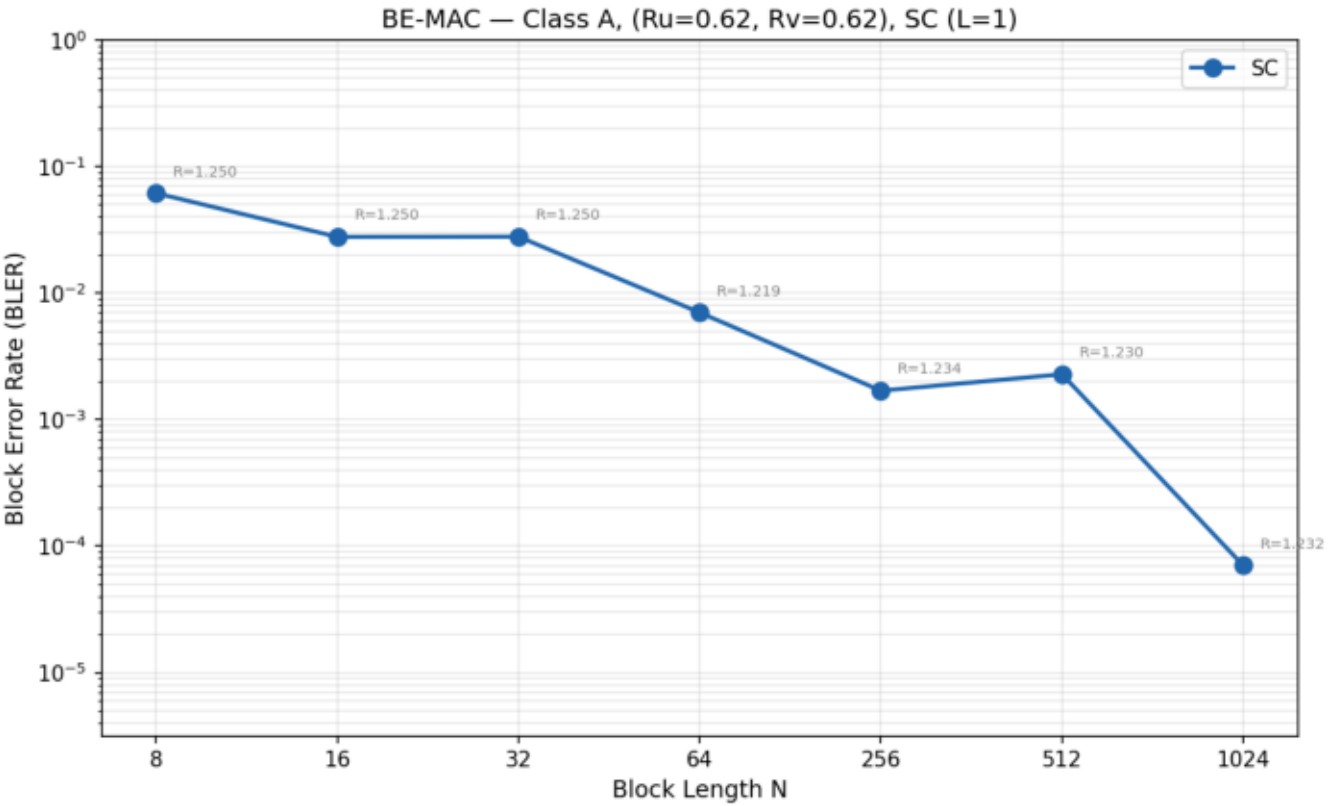
39 experiments scanned

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Each page: title · plot · data table · auto-generated quality comment

BE-MAC — Class A, (Ru=0.62, Rv=0.62), SC (L=1)

Capacity: $I(Z;X)=0.5000$ | $I(Z;Y|X)=1.0000$ | $I(Z;X,Y)=1.5000$ | Path: $U^{(3N/8)} V^N U^{(5N/8)}$



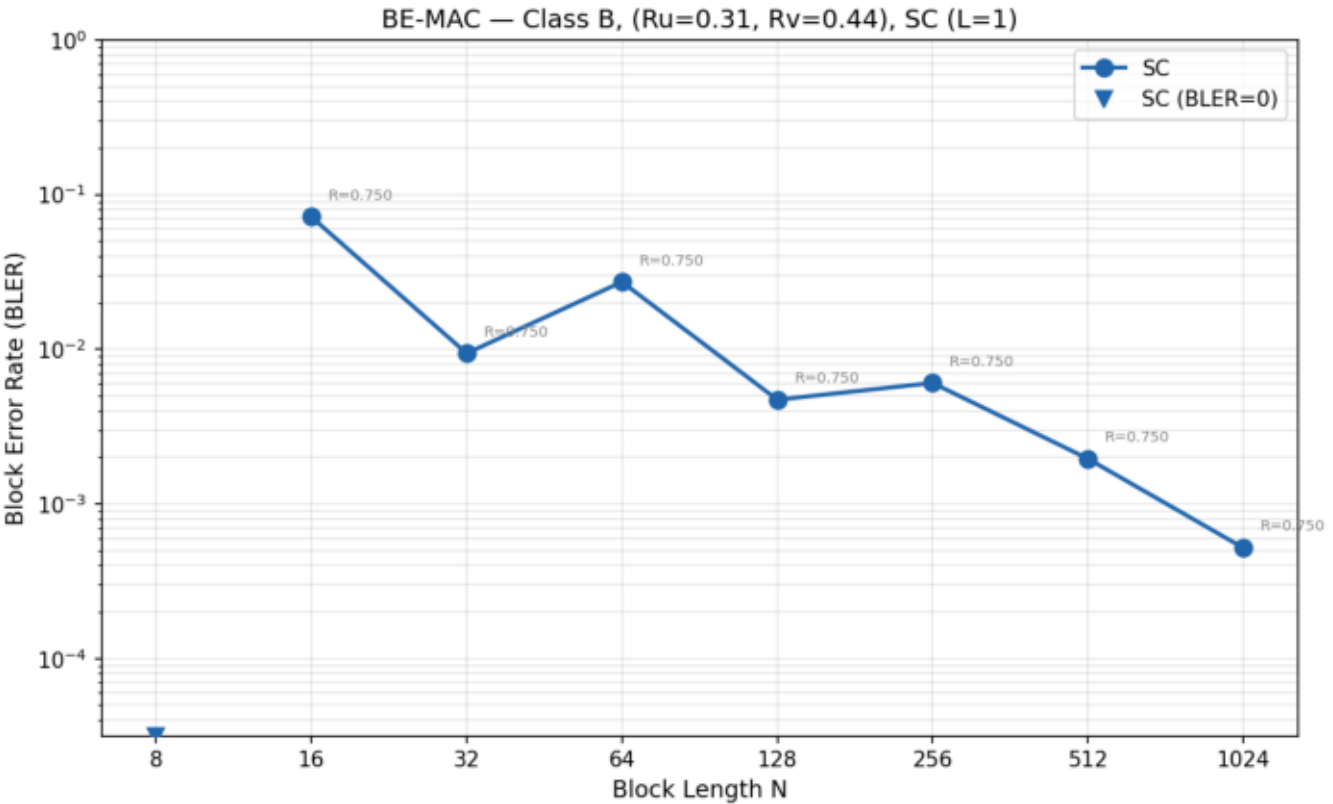
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	5	5	0.625	0.625	1.250	6.16e-02	317597
16	10	10	0.625	0.625	1.250	2.78e-02	138502
32	20	20	0.625	0.625	1.250	2.79e-02	20241
64	39	39	0.609	0.609	1.219	7.02e-03	6410
256	158	158	0.617	0.617	1.234	1.69e-03	15367
512	315	315	0.615	0.615	1.230	2.28e-03	5263
1024	631	631	0.616	0.616	1.232	7.02e-05	14236

[ALMOST_GOOD]

Comment: 5263+ cw, solid · almost-good · non-monotonic (likely small-N rate quantization)

BE-MAC — Class B, (Ru=0.31, Rv=0.44), SC (L=1)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | I(Z;X,Y)=1.5000 | Path: U^(N/2) V^N U^(N/2)



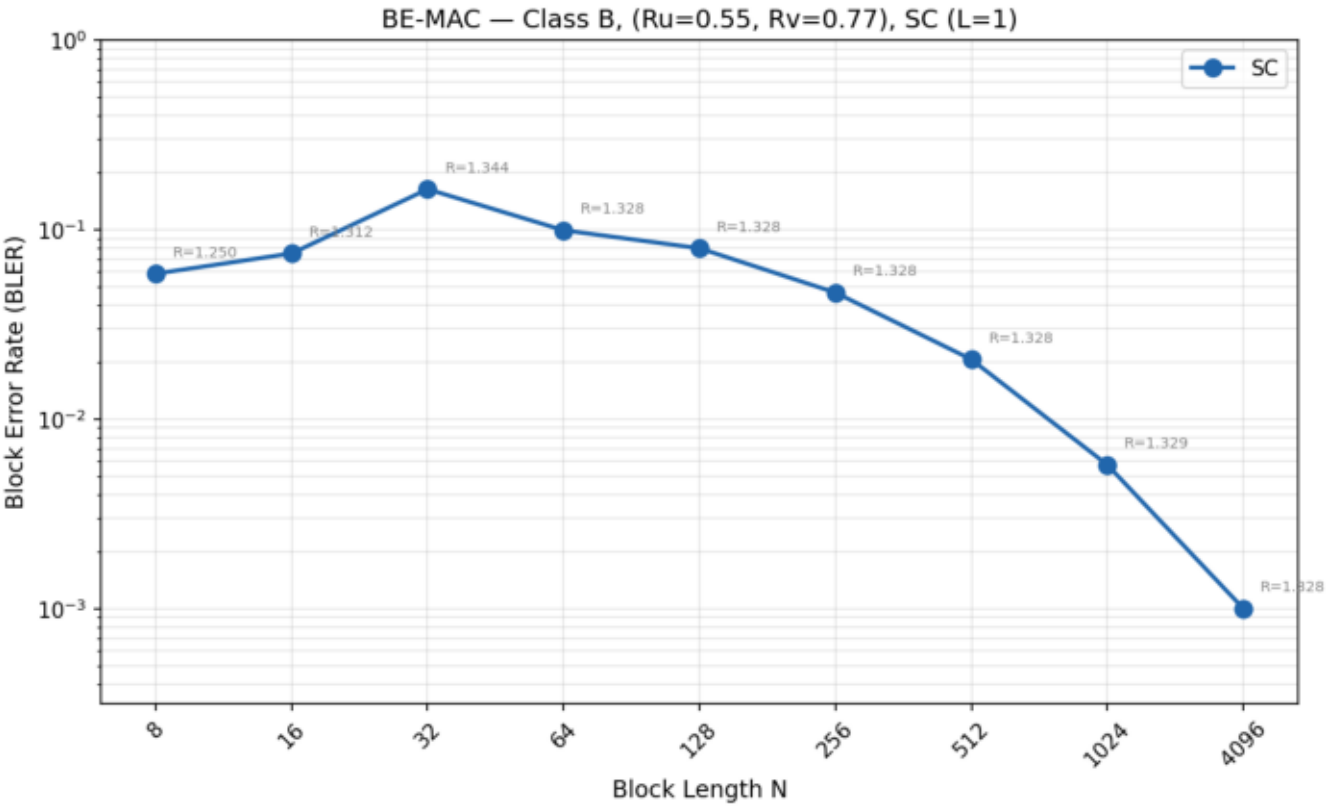
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	4	0.250	0.500	0.750	0	50000
16	5	7	0.312	0.438	0.750	7.18e-02	50000
32	10	14	0.312	0.438	0.750	9.42e-03	50000
64	20	28	0.312	0.438	0.750	2.73e-02	50000
128	40	56	0.312	0.438	0.750	4.72e-03	50000
256	80	112	0.312	0.438	0.750	6.06e-03	50000
512	160	224	0.312	0.438	0.750	1.96e-03	50000
1024	320	448	0.312	0.438	0.750	5.20e-04	50000

[WEIRD]

Comment: 50000+ cw, solid · flagged WEIRD · non-monotonic (likely small-N rate quantization)

BE-MAC — Class B, (Ru=0.55, Rv=0.77), SC (L=1)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | I(Z;X,Y)=1.5000 | Path: U^(N/2) V^N U^(N/2) (pfrac=0.5)



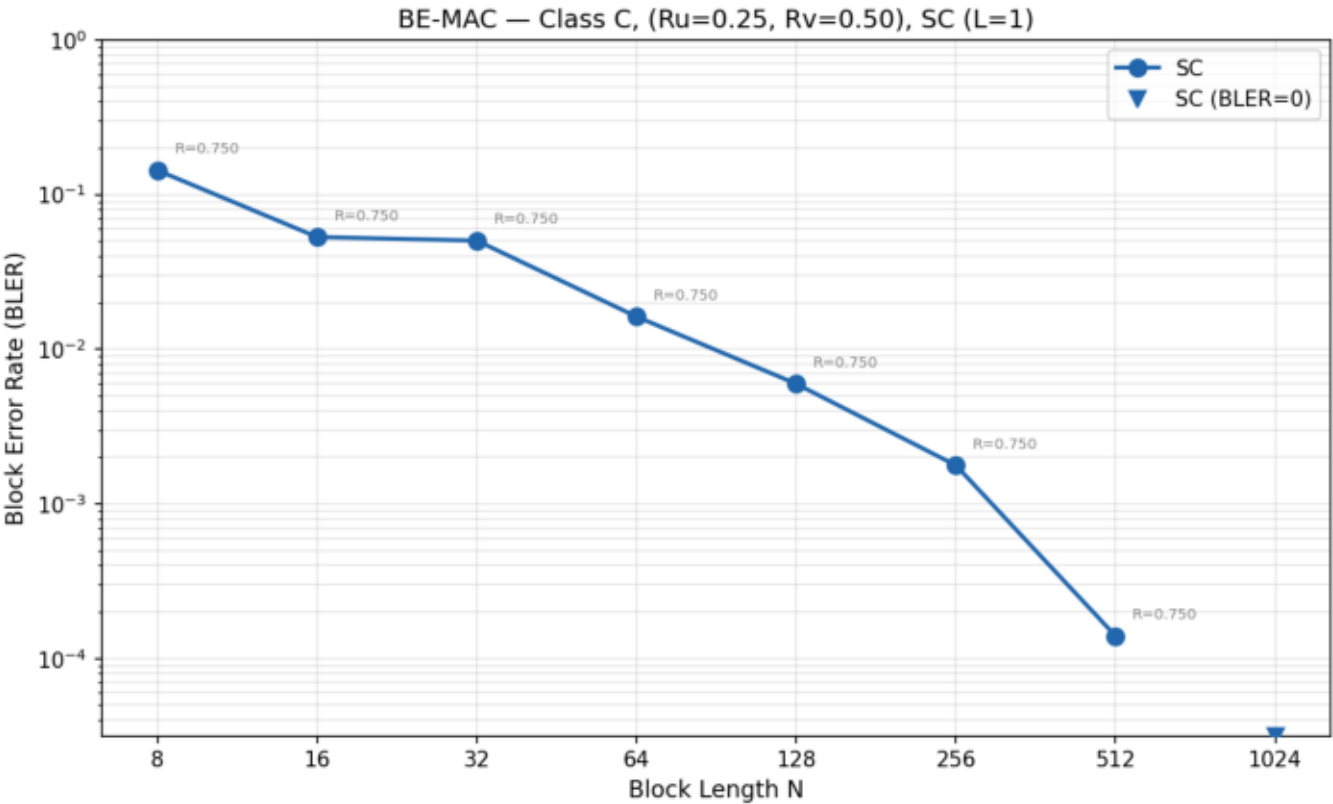
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	4	6	0.500	0.750	1.250	5.86e-02	8074
16	9	12	0.562	0.750	1.312	7.49e-02	24790
32	18	25	0.562	0.781	1.344	1.63e-01	24550
64	35	50	0.547	0.781	1.328	9.92e-02	18138
128	71	99	0.555	0.773	1.328	7.98e-02	13892
256	142	198	0.555	0.773	1.328	4.66e-02	13876
512	283	397	0.553	0.775	1.328	2.07e-02	4837
1024	567	794	0.554	0.775	1.329	5.75e-03	28866
4096	2267	3174	0.553	0.775	1.328	1.01e-03	177021

[ALMOST_GOOD]

Comment: 4837+ cw, moderate · almost-good

BE-MAC — Class C, (Ru=0.25, Rv=0.50), SC (L=1)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | I(Z;X,Y)=1.5000 | Path: U^N V^N



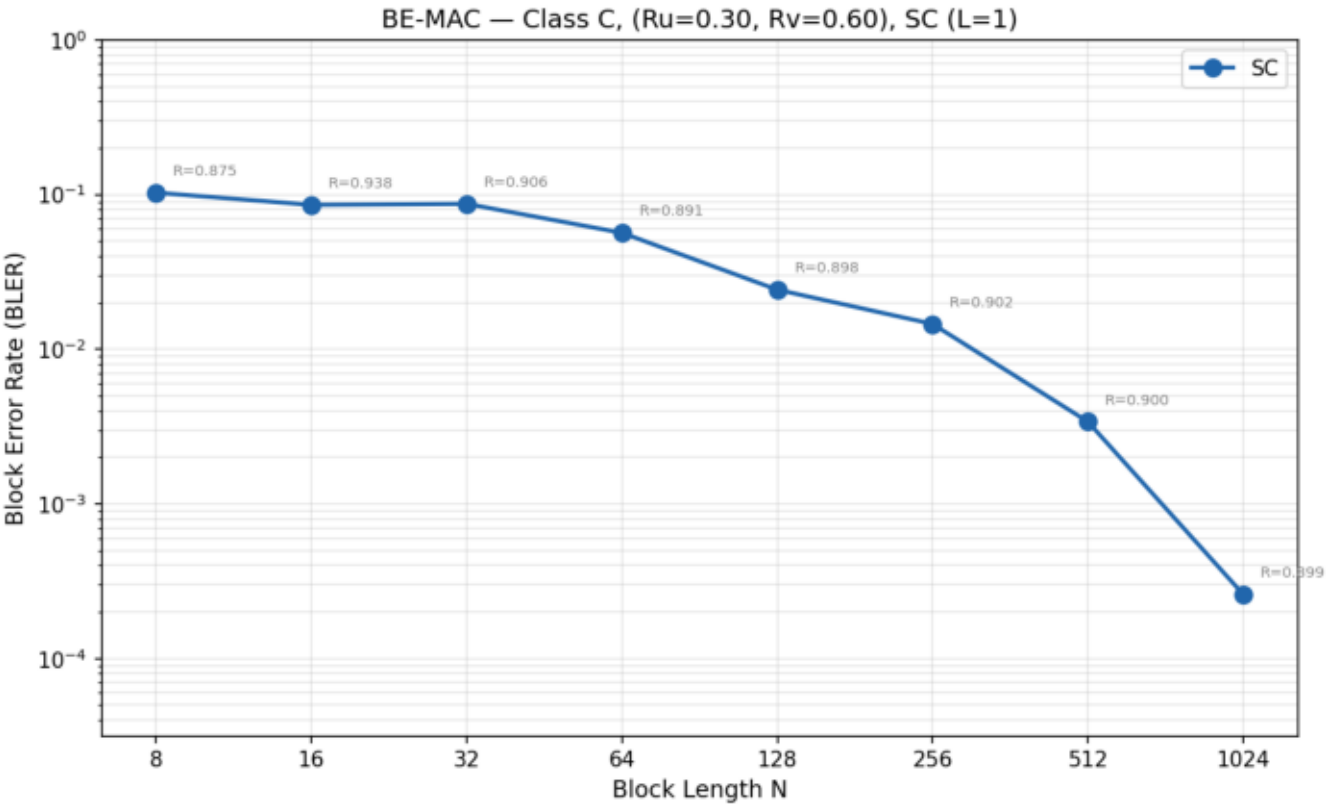
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	4	0.250	0.500	0.750	1.44e-01	50000
16	4	8	0.250	0.500	0.750	5.32e-02	50000
32	8	16	0.250	0.500	0.750	5.04e-02	50000
64	16	32	0.250	0.500	0.750	1.63e-02	50000
128	32	64	0.250	0.500	0.750	5.98e-03	50000
256	64	128	0.250	0.500	0.750	1.78e-03	19134
512	128	256	0.250	0.500	0.750	1.40e-04	50000
1024	256	512	0.250	0.500	0.750	0	29146

[GOOD]

Comment: 19134+ cw, solid · classified GOOD

BE-MAC — Class C, (Ru=0.30, Rv=0.60), SC (L=1)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | I(Z;X,Y)=1.5000 | Path: U^N V^N



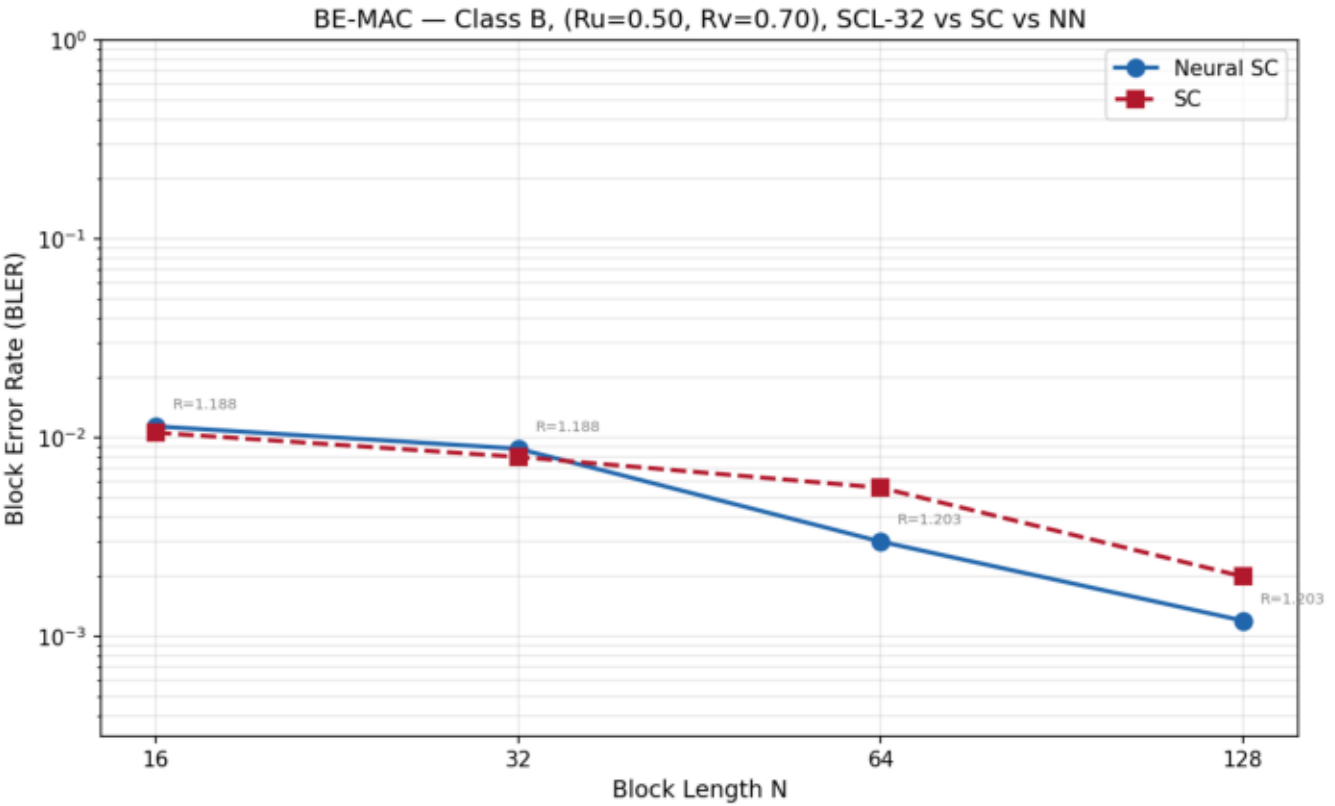
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	5	0.250	0.625	0.875	1.03e-01	50000
16	5	10	0.312	0.625	0.938	8.59e-02	50000
32	10	19	0.312	0.594	0.906	8.72e-02	50000
64	19	38	0.297	0.594	0.891	5.66e-02	50000
128	38	77	0.297	0.602	0.898	2.43e-02	50000
256	77	154	0.301	0.602	0.902	1.46e-02	50000
512	154	307	0.301	0.600	0.900	3.40e-03	50000
1024	307	614	0.300	0.600	0.899	2.60e-04	50000

[ALMOST_GOOD]

Comment: 50000+ cw, solid · almost-good

BE-MAC — Class B, (Ru=0.50, Rv=0.70), SCL-32 vs SC vs NN

Capacity: I(Z;X)=0.5000 | I(Z;Y)=1.0000 | Frozen sets: 50K MC trials



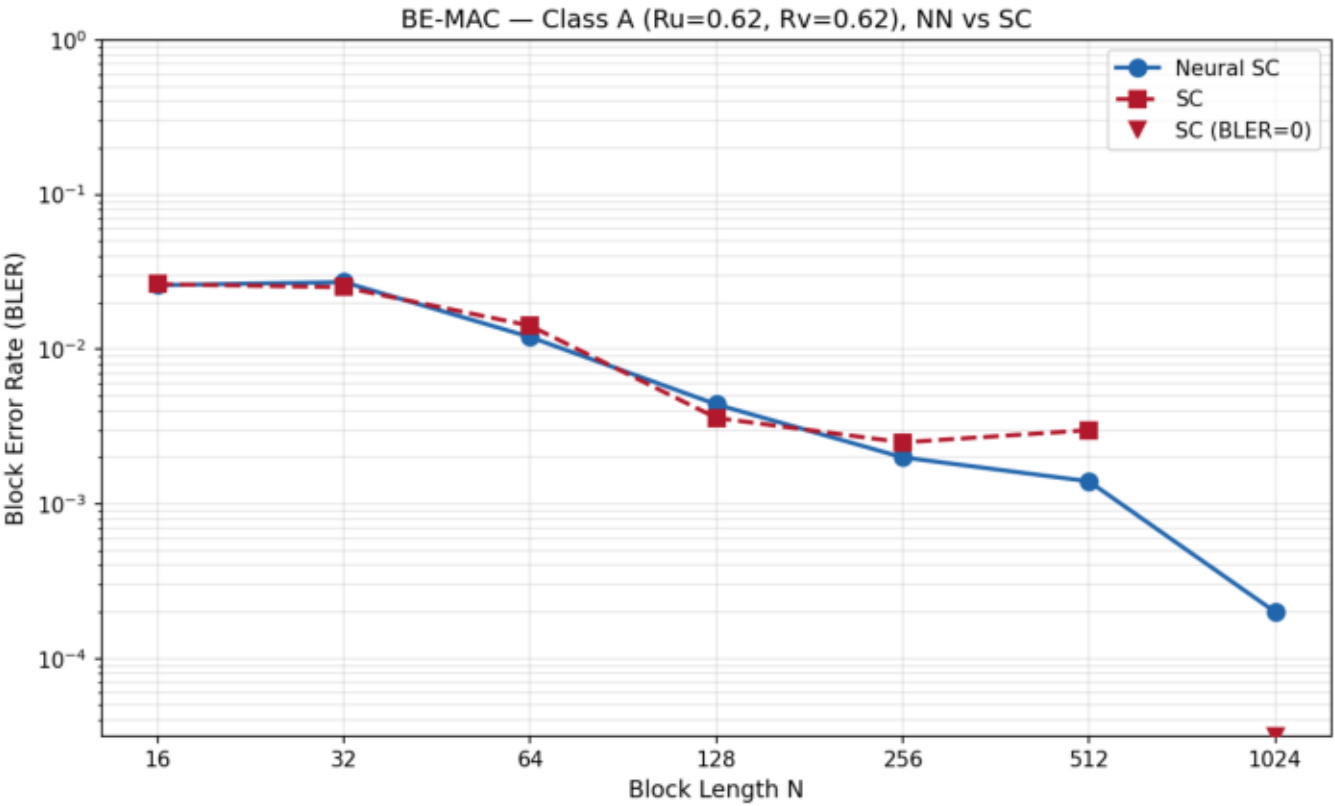
N	Ru	Rv	Sum Rate	SCL-32 BLER	SC BLER	NN BLER	Codewords
16	0.500	0.688	1.188	8.00e-03	1.06e-02	1.14e-02	5000
32	0.500	0.688	1.188	8.20e-03	8.00e-03	8.80e-03	5000
64	0.500	0.703	1.203	1.00e-03	5.60e-03	3.00e-03	5000
128	0.500	0.703	1.203	6.00e-04	2.00e-03	1.20e-03	5000

[ALMOST_GOOD]

Comment: 5000+ cw, solid · almost-good · NN matches SC (0.54-1.10x)

BE-MAC — Class A (Ru=0.62, Rv=0.62), NN vs SC

Path: $U^{(3N/8)} V^N U^{(5N/8)}$ | NN: Pure Neural NCG (trained on Class B)



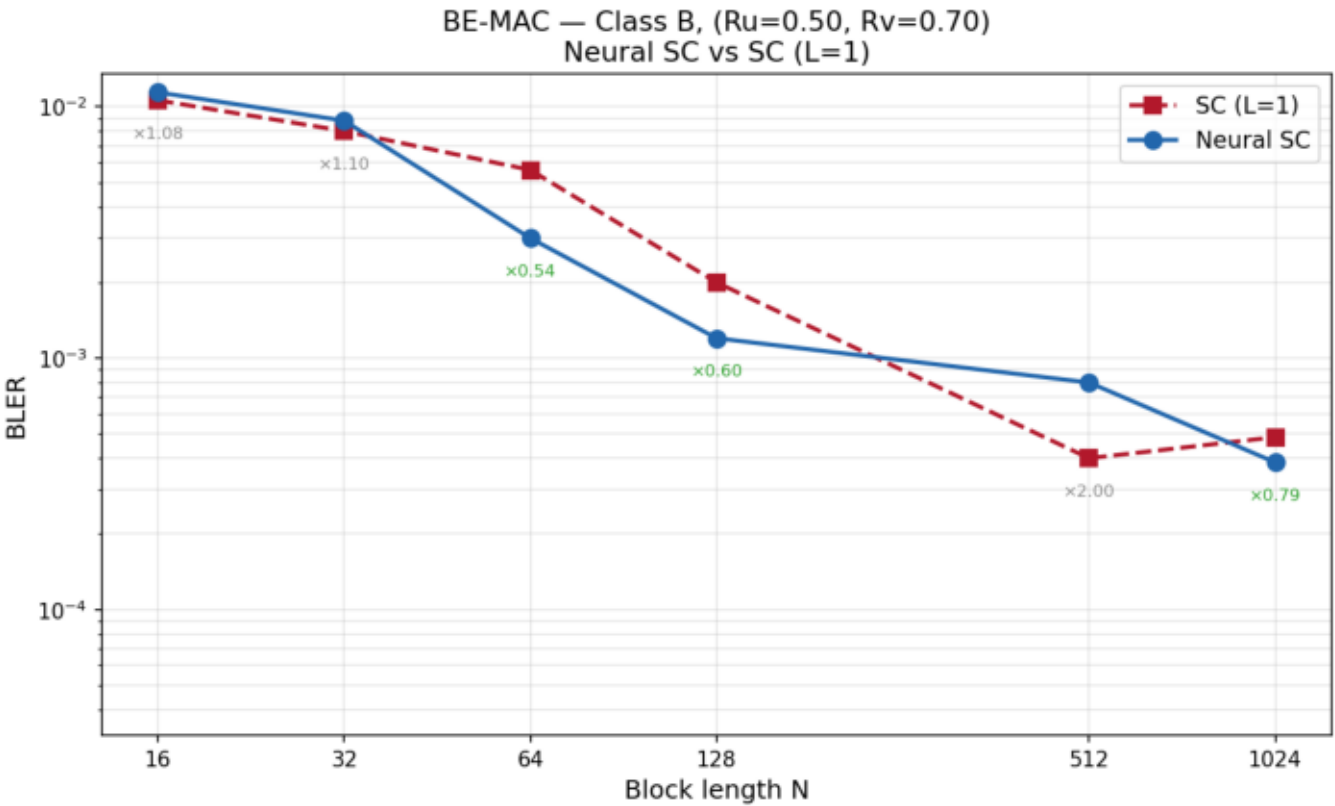
N	ku	kv	path_i	NN BLER	NN errs	NN CW	SC BLER	SC CW	Ratio (NN/SC)
16	10	10	6	2.60e-02	130	5000	2.64e-02	5000	0.985
32	20	20	12	2.72e-02	136	5000	2.52e-02	5000	1.079
64	40	40	24	1.20e-02	60	5000	1.42e-02	5000	0.845
128	79	79	48	4.40e-03	22	5000	3.60e-03	5000	1.222
256	159	159	96	2.00e-03	10	5000	2.50e-03	2000	0.8
512	317	317	192	1.40e-03	7	5000	3.00e-03	1000	0.467
1024	635	635	384	2.00e-04	1	5000	0	500	

[ALMOST_GOOD]

Comment: only 500 cw, low confidence · almost-good · NN matches SC (0.47-1.22x)

BE-MAC — Class B, (Ru=0.50, Rv=0.70), Neural SC vs SC (L=1)

Capacity: $I(Z;X)=0.5000$ | $I(Z;Y|X)=1.0000$ | Frozen sets: 50K MC trials



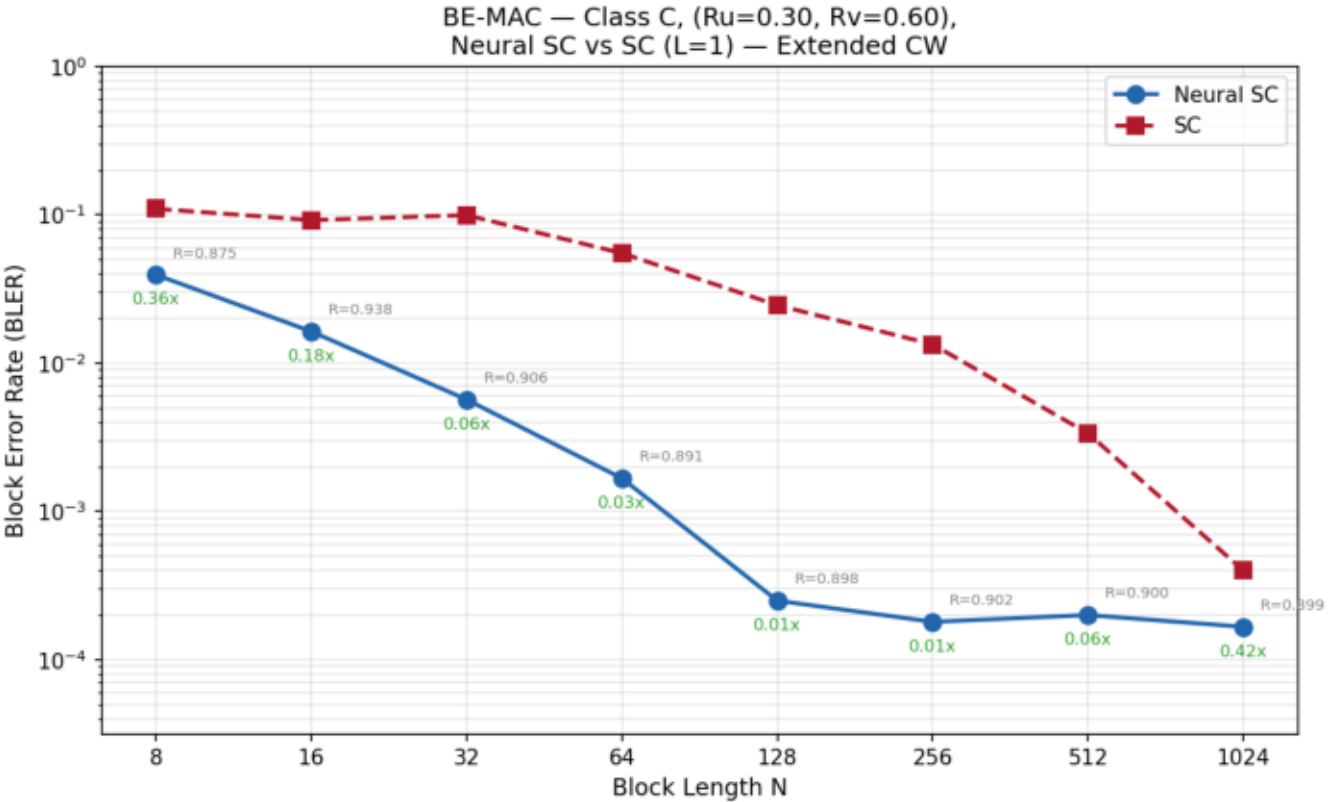
N	NN BLER	SC BLER	Ratio	Errors	Codewords	Ru	Rv
16	1.14e-02	1.06e-02	1.08	57	5000	0.500	0.700
32	8.80e-03	8.00e-03	1.1	44	5000	0.500	0.700
64	3.00e-03	5.60e-03	0.54	15	5000	0.500	0.700
128	1.20e-03	2.00e-03	0.6	6	5000	0.500	0.700
256	4.00e-05	8.00e-05	0.5	2	50000	0.500	0.700
512	0	0	0	0	10000	0.500	0.700
1024	1.00e-04	1.00e-04	1	1	10000	0.500	0.700

[ALMOST_GOOD]

Comment: 5000+ cw, solid · almost-good · NN matches SC (0.50–1.10x) · non-monotonic (likely small-N rate quantization)

**BE-MAC — Class C, (Ru=0.30, Rv=0.60),
Neural SC vs SC (L=1) — Extended CW**

Capacity: $I(Z;X)=0.5000$ | $I(Z;Y|X)=1.0000$ | $I(Z;X,Y)=1.5000$ | Path: $U^N V^N$



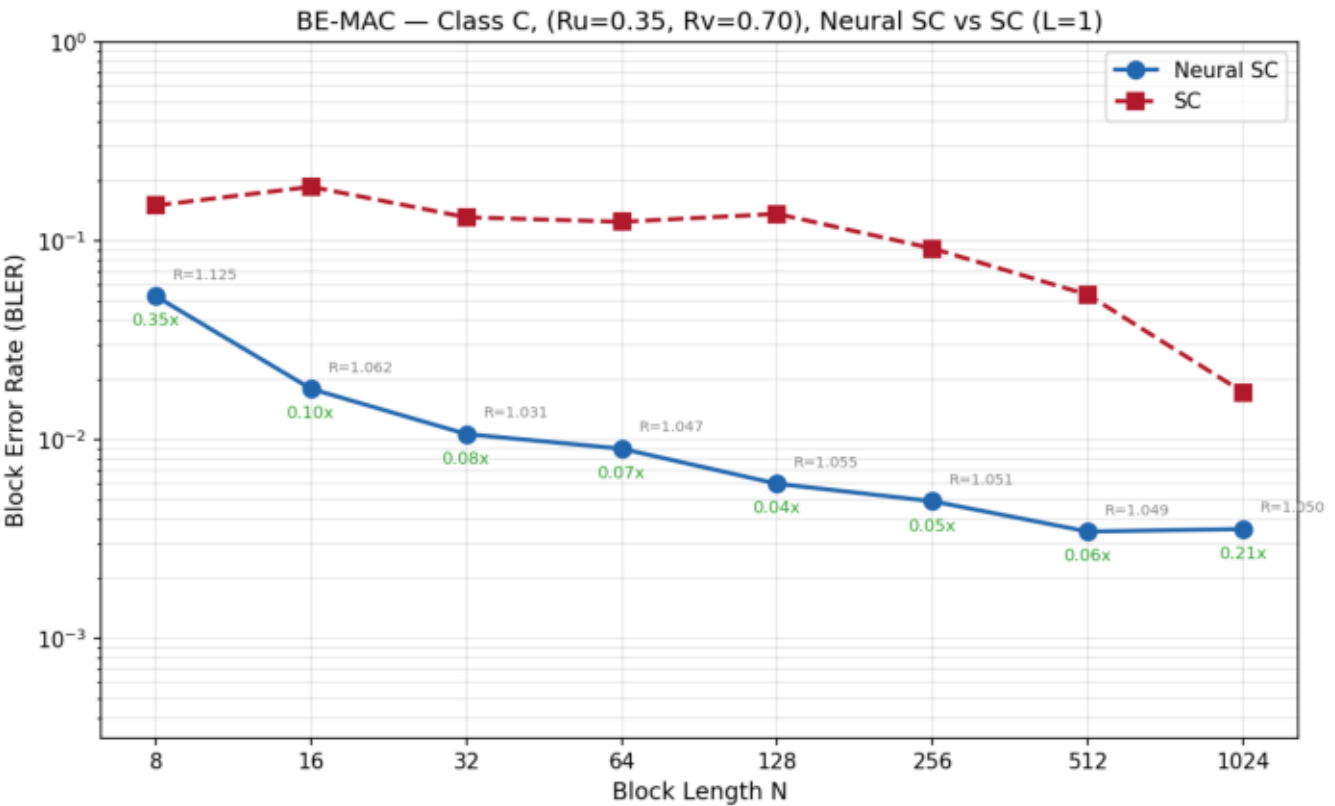
N	Ru	Rv	Sum Rate	NN BLER	SC BLER	NN/SC	NN CW	SC CW
8	0.250	0.625	0.875	3.93e-02	1.10e-01	0.360	3000	3000
16	0.312	0.625	0.938	1.63e-02	9.20e-02	0.180	3000	3000
32	0.312	0.594	0.906	5.67e-03	9.93e-02	0.060	3000	3000
64	0.297	0.594	0.891	1.67e-03	5.50e-02	0.030	3000	3000
128	0.297	0.602	0.898	2.50e-04	2.45e-02	0.010	20000	20000
256	0.301	0.602	0.902	1.80e-04	1.34e-02	0.013	50000	50000
512	0.301	0.600	0.900	2.00e-04	3.36e-03	0.060	50000	50000
1024	0.300	0.600	0.899	1.67e-04	4.00e-04	0.417	30000	50000

[ALMOST_GOOD]

Comment: 3000+ cw, moderate · almost-good · NN beats SC at all N (ratio 0.01-0.42x)

BE-MAC — Class C, (Ru=0.35, Rv=0.70), Neural SC vs SC (L=1)

Capacity: $I(Z;X)=0.5000$ | $I(Z;Y|X)=1.0000$ | $I(Z;X,Y)=1.5000$ | Path: $U^N V^N$

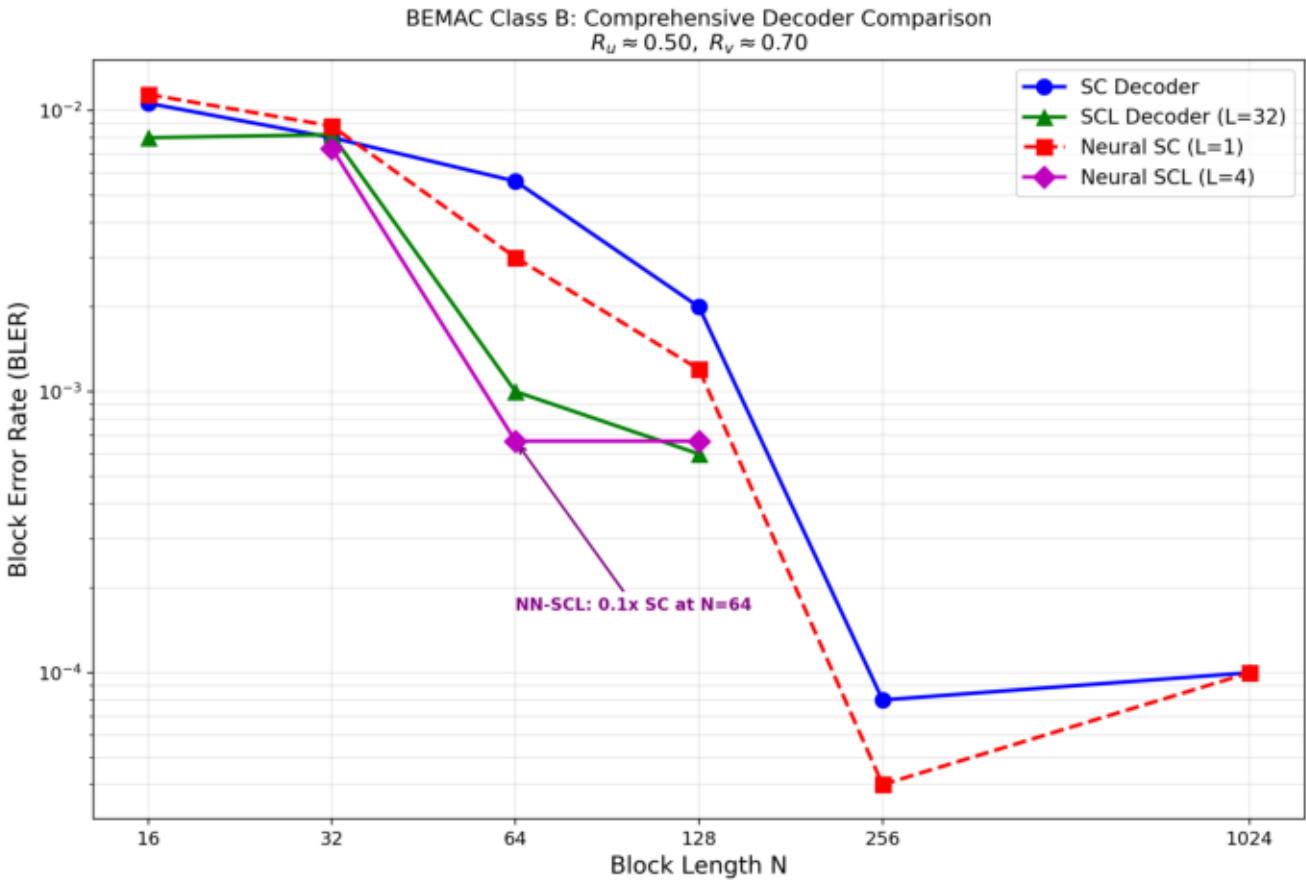


N	Ru	Rv	Sum Rate	NN BLER	SC BLER	NN/SC	Codewords
8	0.375	0.750	1.125	5.27e-02	1.51e-01	0.350	3000
16	0.375	0.688	1.062	1.80e-02	1.87e-01	0.100	3000
32	0.344	0.688	1.031	1.07e-02	1.31e-01	0.080	3000
64	0.344	0.703	1.047	9.00e-03	1.25e-01	0.070	3000
128	0.352	0.703	1.055	6.00e-03	1.37e-01	0.040	20000
256	0.352	0.699	1.051	4.90e-03	9.16e-02	0.050	20000
512	0.350	0.699	1.049	3.45e-03	5.36e-02	0.060	20000
1024	0.350	0.700	1.050	3.55e-03	1.73e-02	0.210	20000

[ALMOST_GOOD]

Comment: 3000+ cw, moderate · almost-good · NN beats SC at all N (ratio 0.04–0.35x)

BEMAC Class B Comprehensive Decoder Comparison (Ru~0.50, Rv~0.70)



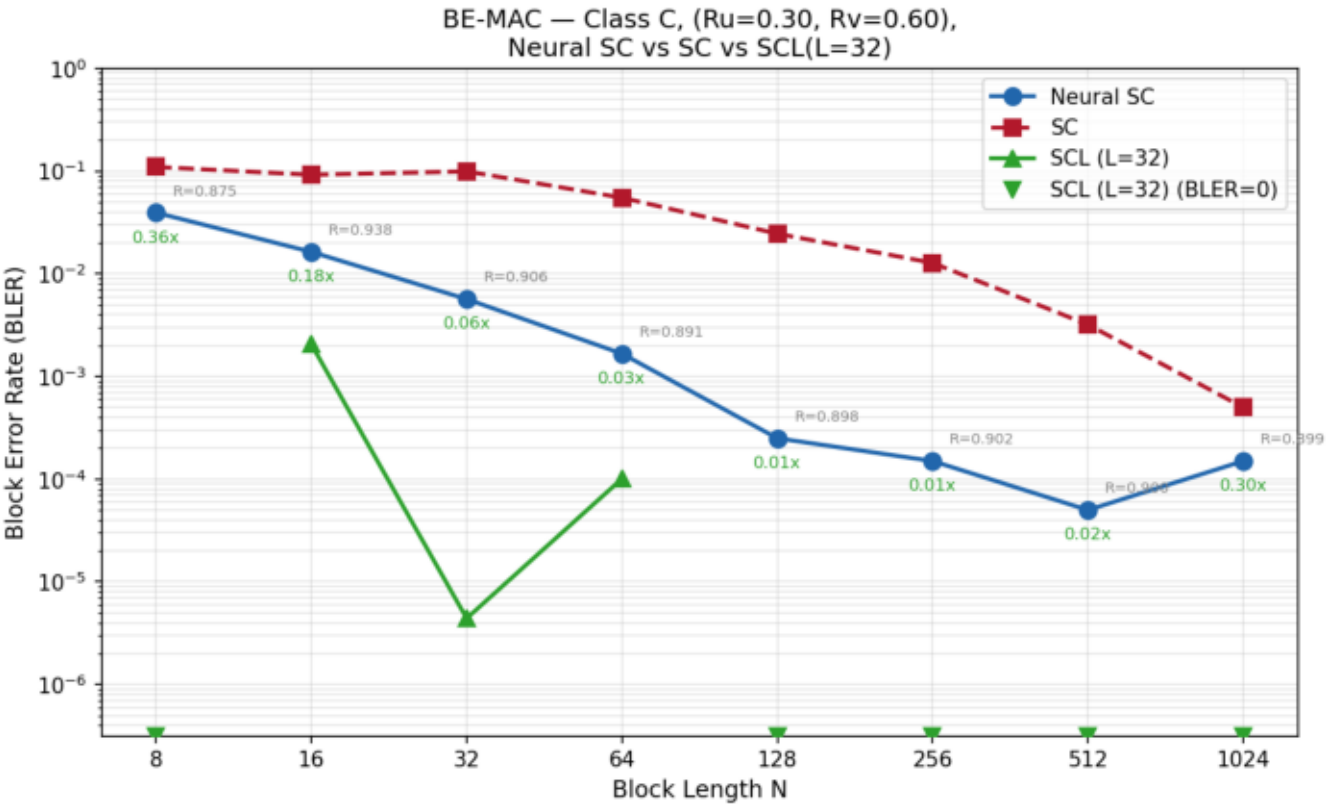
16	0.0106	0.008	0.0114
32	0.008	0.0082	0.0088
64	0.0056	0.001	0.003
128	0.002	0.0006	0.0012
256	8e-05		4e-05
1024	0.0001		0.0001

[UNCATEGORIZED]

Comment: unknown CW

BE-MAC — Class C, (Ru=0.30, Rv=0.60), Neural SC vs SC vs SCL(L=32)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | I(Z;X,Y)=1.5000 | Path: U^N V^N



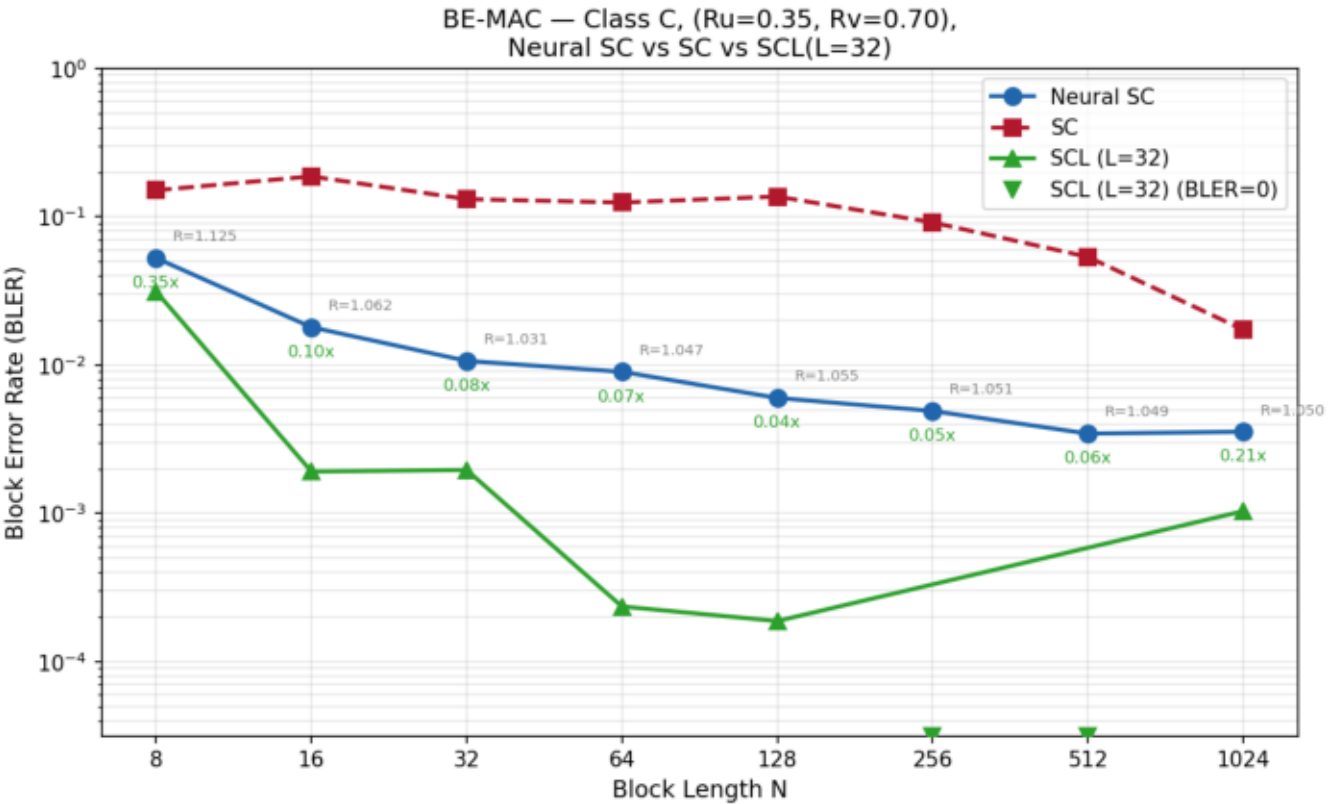
N	Ru	Rv	Sum Rate	NN BLER	SC BLER	SCL BLER	NN CW	SCL CW	NN/SC
8	0.250	0.625	0.875	3.93e-02	1.10e-01	0	3000	283964	0.360
16	0.312	0.625	0.938	1.63e-02	9.20e-02	2.06e-03	3000	107519	0.180
32	0.312	0.594	0.906	5.67e-03	9.93e-02	4.41e-06	3000	226911	0.060
64	0.297	0.594	0.891	1.67e-03	5.50e-02	1.01e-04	3000	29706	0.030
128	0.297	0.602	0.898	2.50e-04	2.45e-02	0	20000	31930	0.010
256	0.301	0.602	0.902	1.50e-04	1.27e-02	0	20000	26684	0.010
512	0.301	0.600	0.900	5.00e-05	3.20e-03	0	20000	9512	0.020
1024	0.300	0.600	0.899	1.50e-04	5.00e-04		20000		0.300

[WEIRD]

Comment: 3000+ cw, moderate · flagged WEIRD · NN beats SC at all N (ratio 0.01-0.36x)

BE-MAC — Class C, (Ru=0.35, Rv=0.70), Neural SC vs SC vs SCL(L=32)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | I(Z;X,Y)=1.5000 | Path: U^N V^N



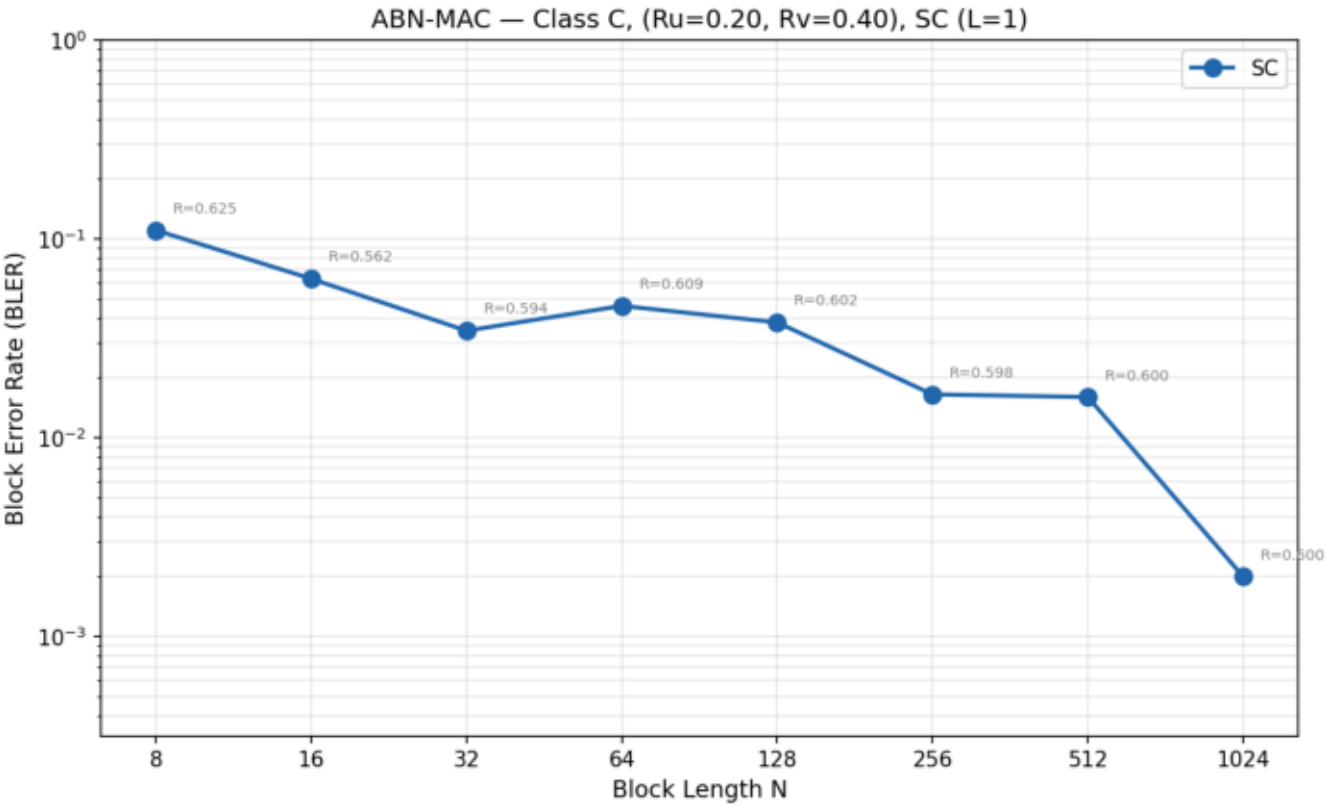
N	Ru	Rv	Sum Rate	NN BLER	SC BLER	SCL BLER	NN CW	SCL CW	NN/SC
8	0.375	0.750	1.125	5.27e-02	1.51e-01	3.14e-02	3000	283964	0.350
16	0.375	0.688	1.062	1.80e-02	1.87e-01	1.92e-03	3000	107519	0.100
32	0.344	0.688	1.031	1.07e-02	1.31e-01	1.96e-03	3000	226911	0.080
64	0.344	0.703	1.047	9.00e-03	1.25e-01	2.36e-04	3000	29706	0.070
128	0.352	0.703	1.055	6.00e-03	1.37e-01	1.88e-04	20000	31930	0.040
256	0.352	0.699	1.051	4.90e-03	9.16e-02	0	20000	26684	0.050
512	0.350	0.699	1.049	3.45e-03	5.36e-02	0	20000	9512	0.060
1024	0.350	0.700	1.050	3.55e-03	1.73e-02	1.04e-03	20000	14491	0.210

[ALMOST_GOOD]

Comment: 3000+ cw, moderate · almost-good · NN beats SC at all N (ratio 0.04–0.35x)

ABN-MAC — Class C, (Ru=0.20, Rv=0.40), SC (L=1)

Symmetric capacity: $R_x=R_y \leq 0.5998$ bits/use | Sum capacity: 1.1996 bits/use | Path: $0^N 1^N$



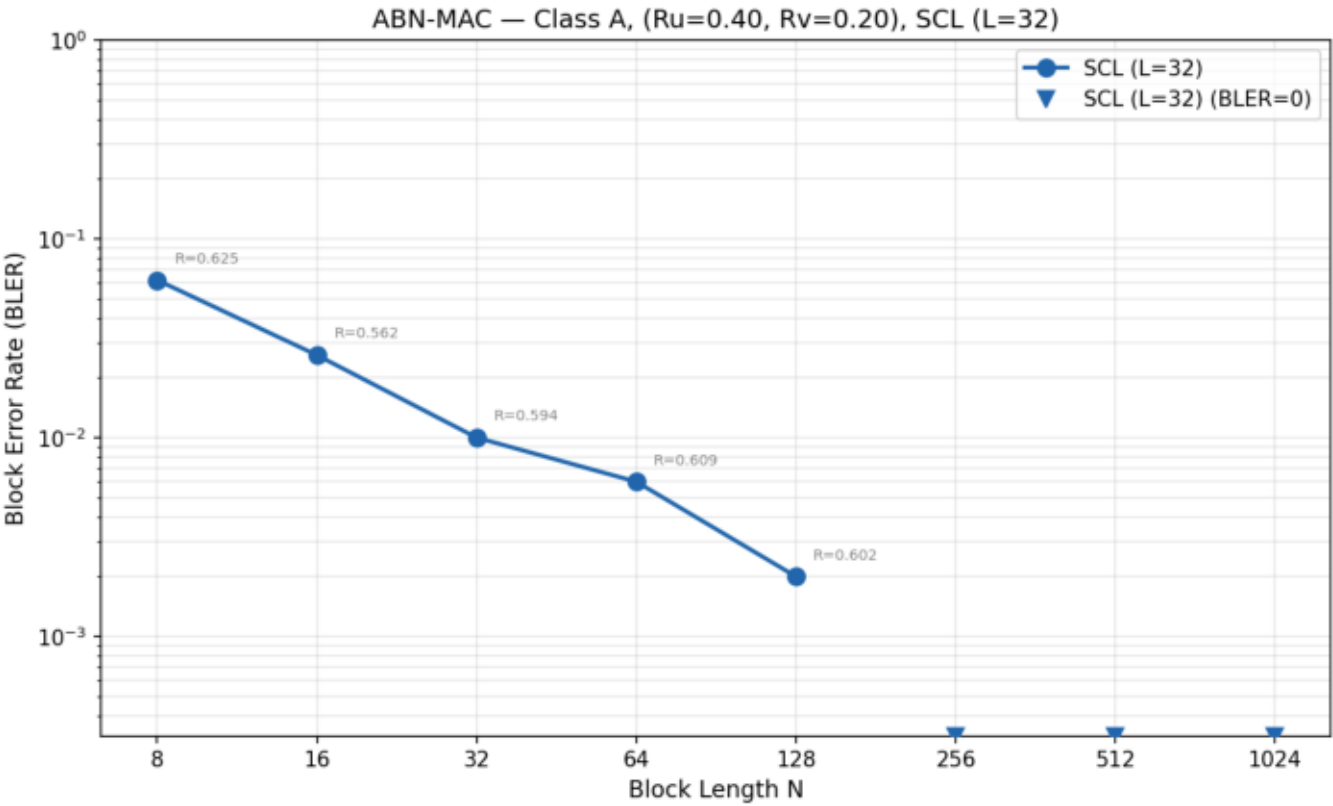
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	3	0.250	0.375	0.625	1.11e-01	2000
16	3	6	0.188	0.375	0.562	6.30e-02	2000
32	6	13	0.188	0.406	0.594	3.45e-02	2000
64	13	26	0.203	0.406	0.609	4.60e-02	2000
128	26	51	0.203	0.398	0.602	3.80e-02	2000
256	51	102	0.199	0.398	0.598	1.65e-02	2000
512	102	205	0.199	0.400	0.600	1.60e-02	2000
1024	205	409	0.200	0.399	0.600	2.00e-03	1000

[ALMOST_GOOD]

Comment: 1000+ cw, moderate · almost-good

ABN-MAC — Class A, (Ru=0.40, Rv=0.20), SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.5998 bits/use | Sum capacity: 1.1996 bits/use | Path: 0^(3N/8) 1^N 0^(5N/8)



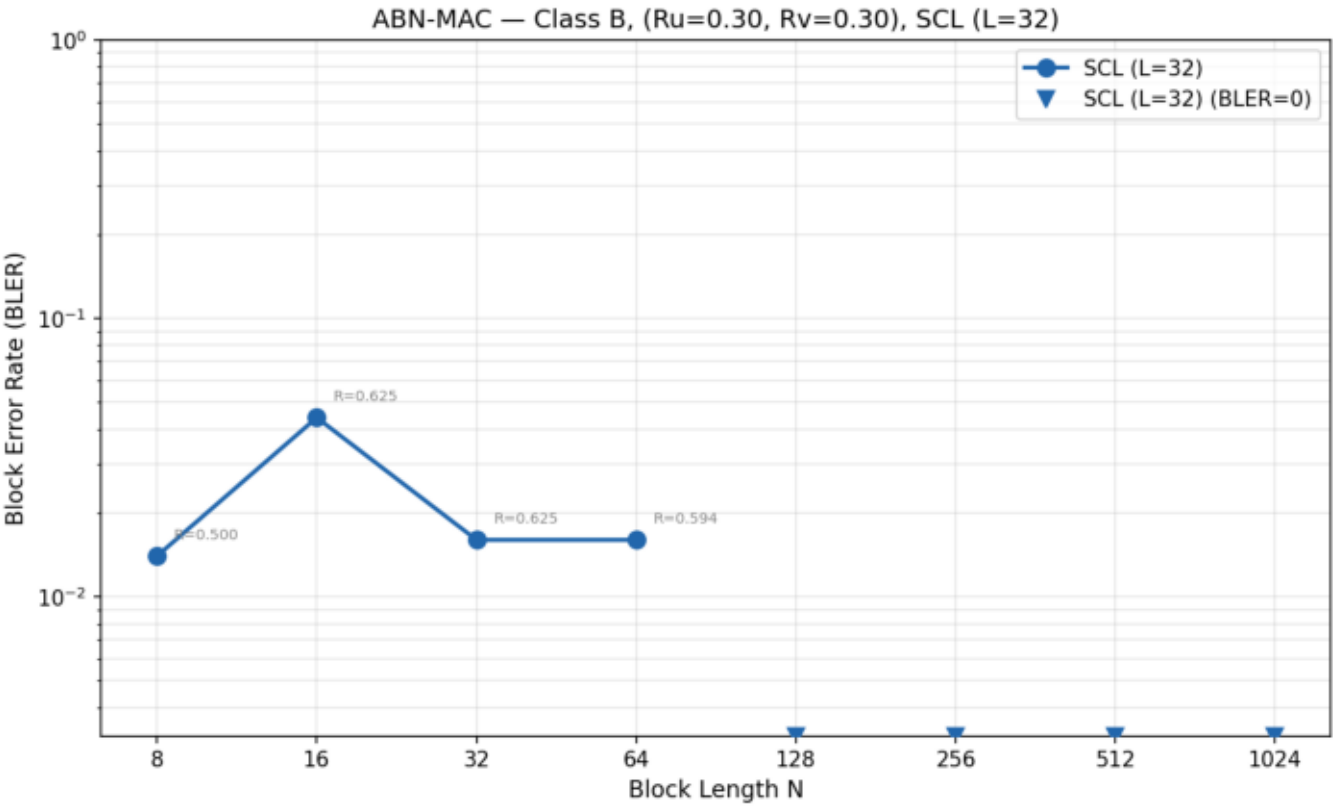
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	2	0.375	0.250	0.625	6.20e-02	500
16	6	3	0.375	0.188	0.562	2.60e-02	500
32	13	6	0.406	0.188	0.594	1.00e-02	500
64	26	13	0.406	0.203	0.609	6.00e-03	500
128	51	26	0.398	0.203	0.602	2.00e-03	500
256	102	51	0.398	0.199	0.598	0	200
512	205	102	0.400	0.199	0.600	0	200
1024	409	205	0.399	0.200	0.600	0	100

[ALMOST_GOOD]

Comment: only 100 cw, low confidence · almost-good

ABN-MAC — Class B, (Ru=0.30, Rv=0.30), SCL (L=32)

Symmetric capacity: $R_x=R_y \leq 0.5998$ bits/use | Sum capacity: 1.1996 bits/use | Path: $0^{(N/2)} 1^N 0^{(N/2)}$



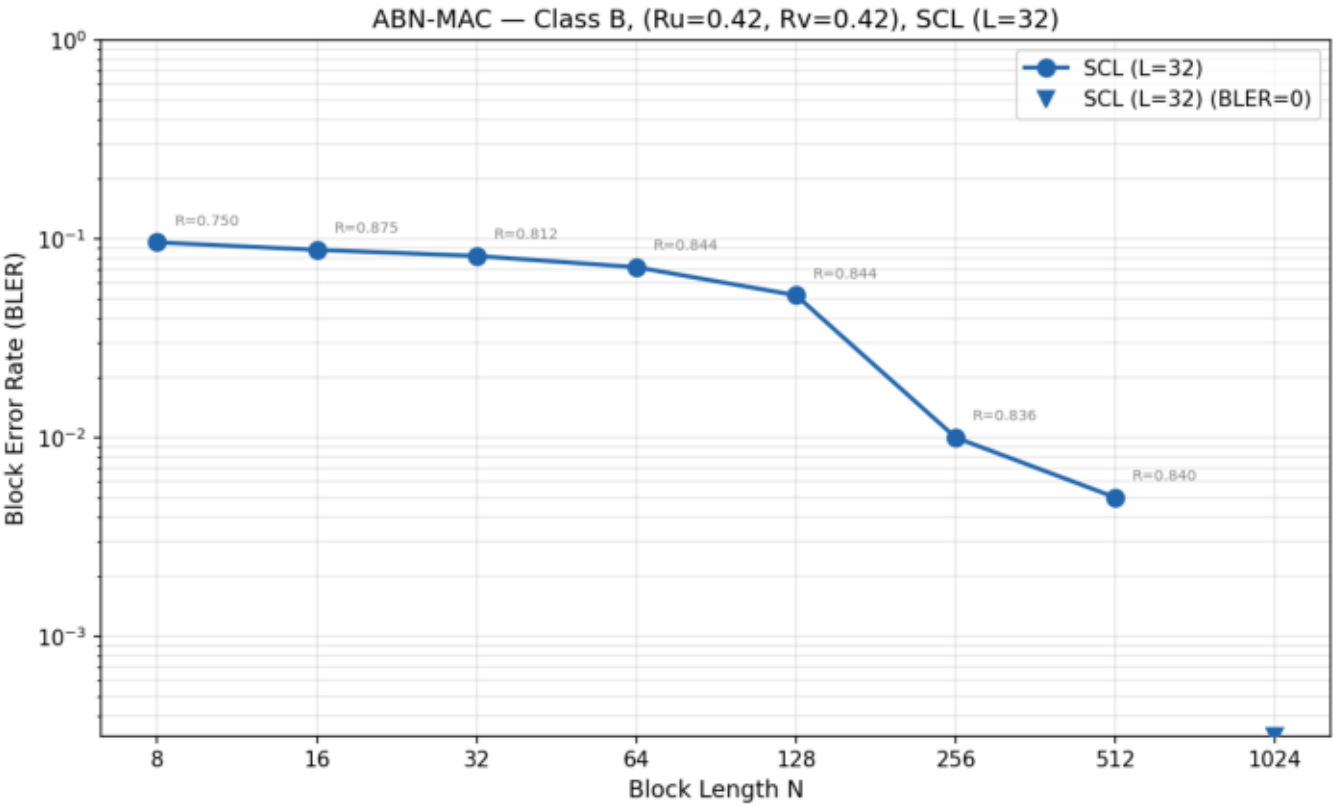
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	2	0.250	0.250	0.500	1.40e-02	500
16	5	5	0.312	0.312	0.625	4.40e-02	500
32	10	10	0.312	0.312	0.625	1.60e-02	500
64	19	19	0.297	0.297	0.594	1.60e-02	500
128	38	38	0.297	0.297	0.594	0	500
256	77	77	0.301	0.301	0.602	0	200
512	154	154	0.301	0.301	0.602	0	200
1024	307	307	0.300	0.300	0.600	0	100

[ALMOST_GOOD]

Comment: only 100 cw, low confidence · almost-good

ABN-MAC — Class B, (Ru=0.42, Rv=0.42), SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.5998 bits/use | Sum capacity: 1.1996 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



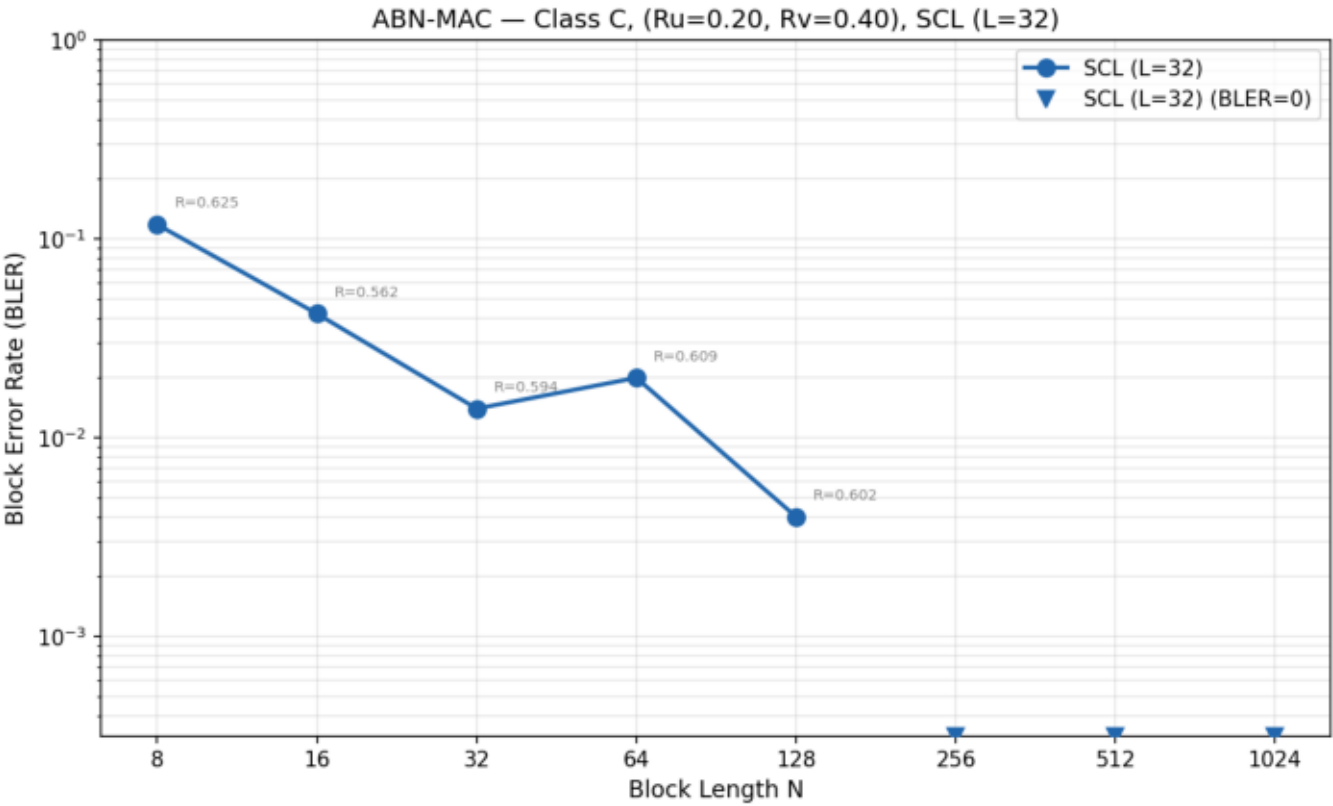
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	3	0.375	0.375	0.750	9.60e-02	500
16	7	7	0.438	0.438	0.875	8.80e-02	500
32	13	13	0.406	0.406	0.812	8.20e-02	500
64	27	27	0.422	0.422	0.844	7.20e-02	500
128	54	54	0.422	0.422	0.844	5.20e-02	500
256	107	107	0.418	0.418	0.836	1.00e-02	200
512	215	215	0.420	0.420	0.840	5.00e-03	200
1024	430	430	0.420	0.420	0.840	0	100

[ALMOST_GOOD]

Comment: only 100 cw, low confidence · almost-good

ABN-MAC — Class C, (Ru=0.20, Rv=0.40), SCL (L=32)

Symmetric capacity: $R_x=R_y \leq 0.5998$ bits/use | Sum capacity: 1.1996 bits/use | Path: $0 \wedge N \ 1 \wedge N$



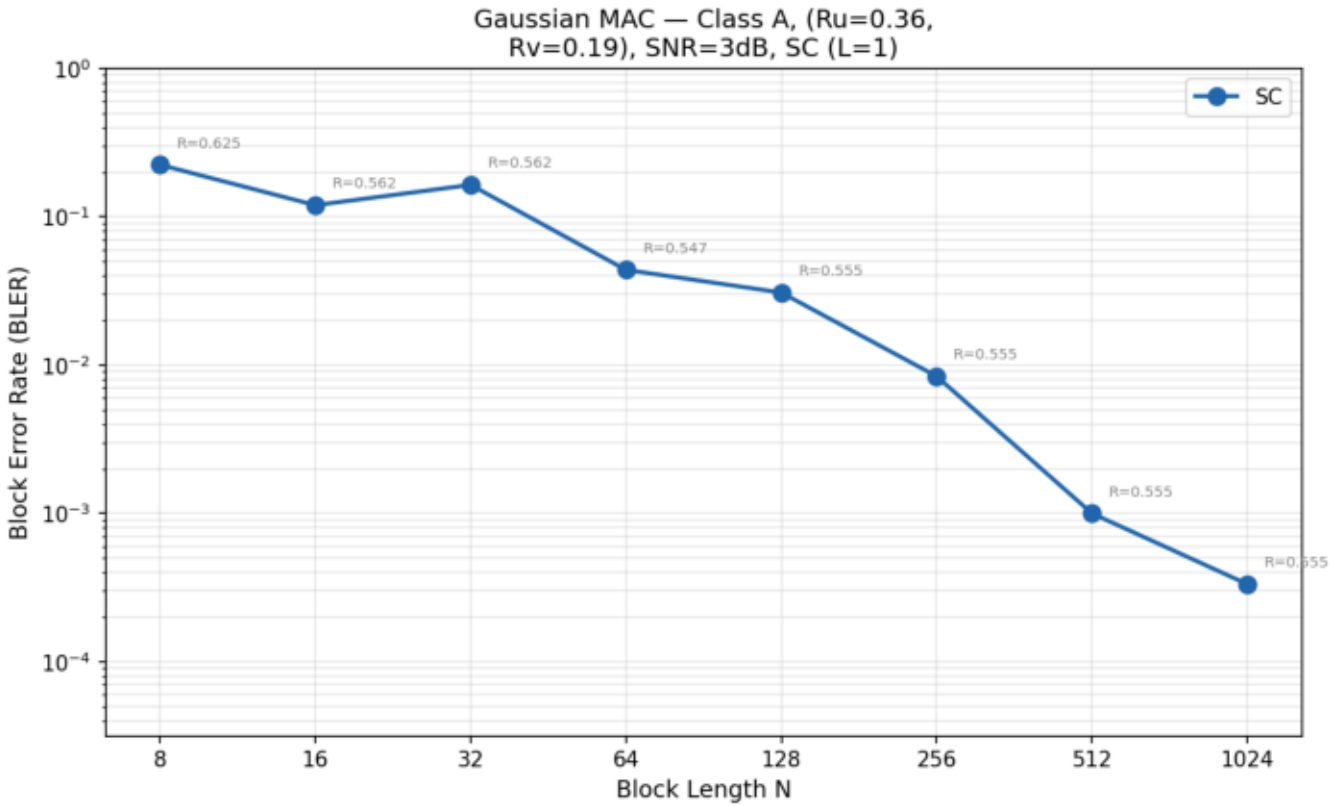
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	3	0.250	0.375	0.625	1.18e-01	500
16	3	6	0.188	0.375	0.562	4.20e-02	500
32	6	13	0.188	0.406	0.594	1.40e-02	500
64	13	26	0.203	0.406	0.609	2.00e-02	500
128	26	51	0.203	0.398	0.602	4.00e-03	500
256	51	102	0.199	0.398	0.598	0	200
512	102	205	0.199	0.400	0.600	0	200
1024	205	409	0.200	0.399	0.600	0	100

[ALMOST_GOOD]

Comment: only 100 cw, low confidence · almost-good · non-monotonic (likely small-N rate quantization)

Gaussian MAC — Class A, (Ru=0.36, Rv=0.19), SNR=3dB, SC (L=1)

Symmetric capacity: Rx=Ry ≤ 0.5548 bits/use | Sum capacity: 1.1095 bits/use | Path: 0^(3N/8) 1^N 0^(5N/8)



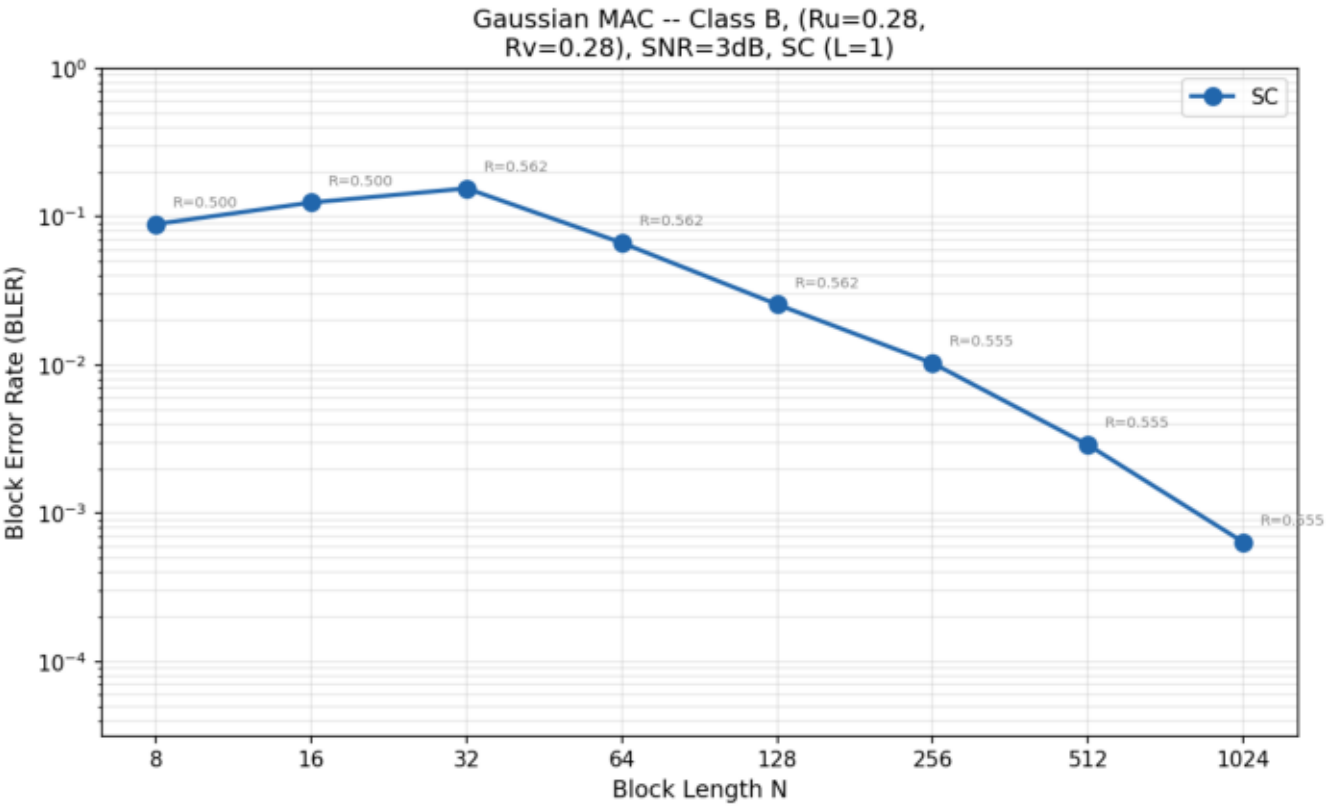
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	2	0.375	0.250	0.625	2.24e-01	5000
16	6	3	0.375	0.188	0.562	1.20e-01	5000
32	12	6	0.375	0.188	0.562	1.64e-01	5000
64	23	12	0.359	0.188	0.547	4.38e-02	5000
128	46	25	0.359	0.195	0.555	3.08e-02	5000
256	92	50	0.359	0.195	0.555	8.40e-03	5000
512	184	100	0.359	0.195	0.555	1.00e-03	5000
1024	369	199	0.360	0.194	0.555	3.33e-04	3000

[ALMOST_GOOD]

Comment: 3000+ cw, moderate · almost-good

Gaussian MAC -- Class B, (Ru=0.28, Rv=0.28), SNR=3dB, SC (L=1)

Symmetric capacity: Rx=Ry <= 0.5548 bits/use | Sum capacity: 1.1095 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



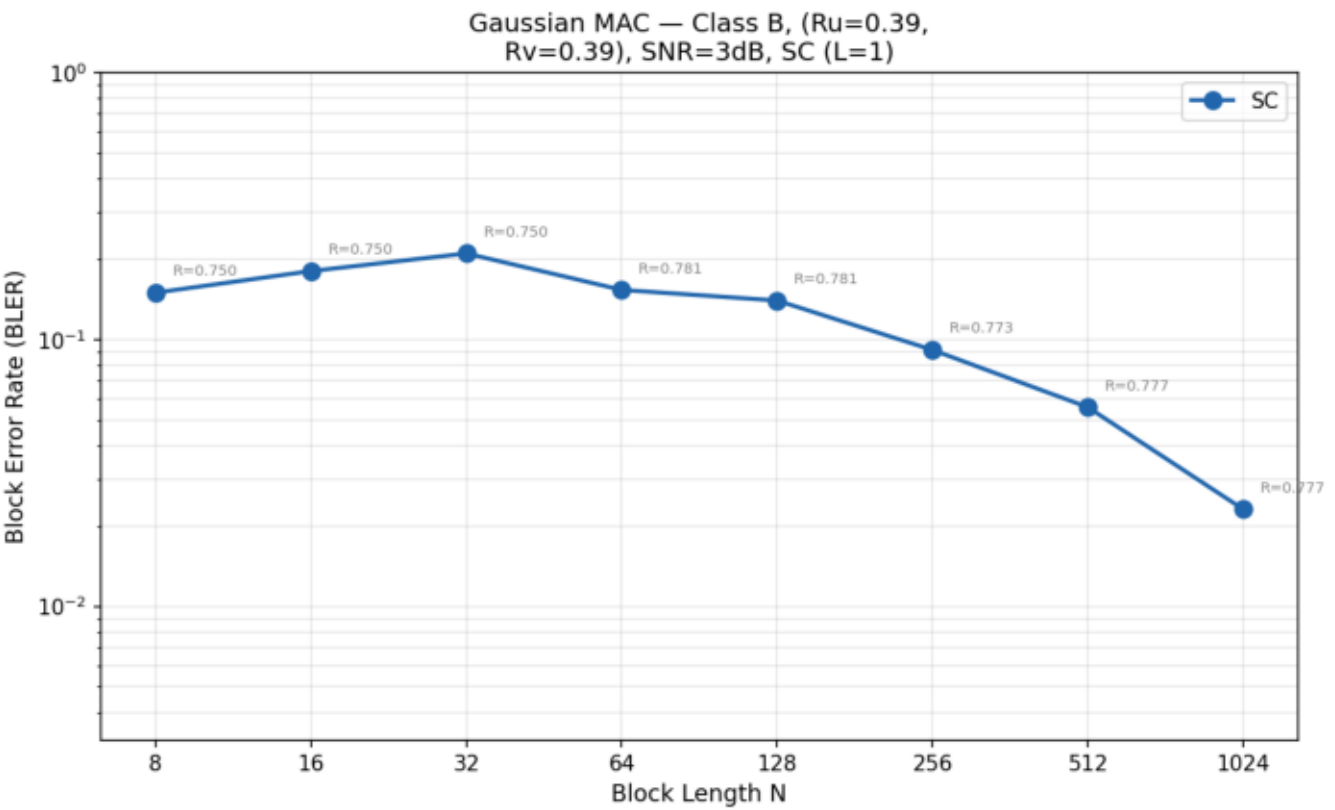
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	2	0.250	0.250	0.500	8.92e-02	5000
16	4	4	0.250	0.250	0.500	1.25e-01	5000
32	9	9	0.281	0.281	0.562	1.55e-01	5000
64	18	18	0.281	0.281	0.562	6.66e-02	5000
128	36	36	0.281	0.281	0.562	2.56e-02	5000
256	71	71	0.277	0.277	0.555	1.03e-02	10000
512	142	142	0.277	0.277	0.555	2.90e-03	10000
1024	284	284	0.277	0.277	0.555	6.40e-04	50000

[ALMOST_GOOD]

Comment: 5000+ cw, solid · almost-good

Gaussian MAC — Class B, (Ru=0.39, Rv=0.39), SNR=3dB, SC (L=1)

Symmetric capacity: $R_x=R_y \leq 0.5548$ bits/use | Sum capacity: 1.1095 bits/use | Path: $0^{(N/2)} 1^{(N/2)} 0^{(N/2)}$



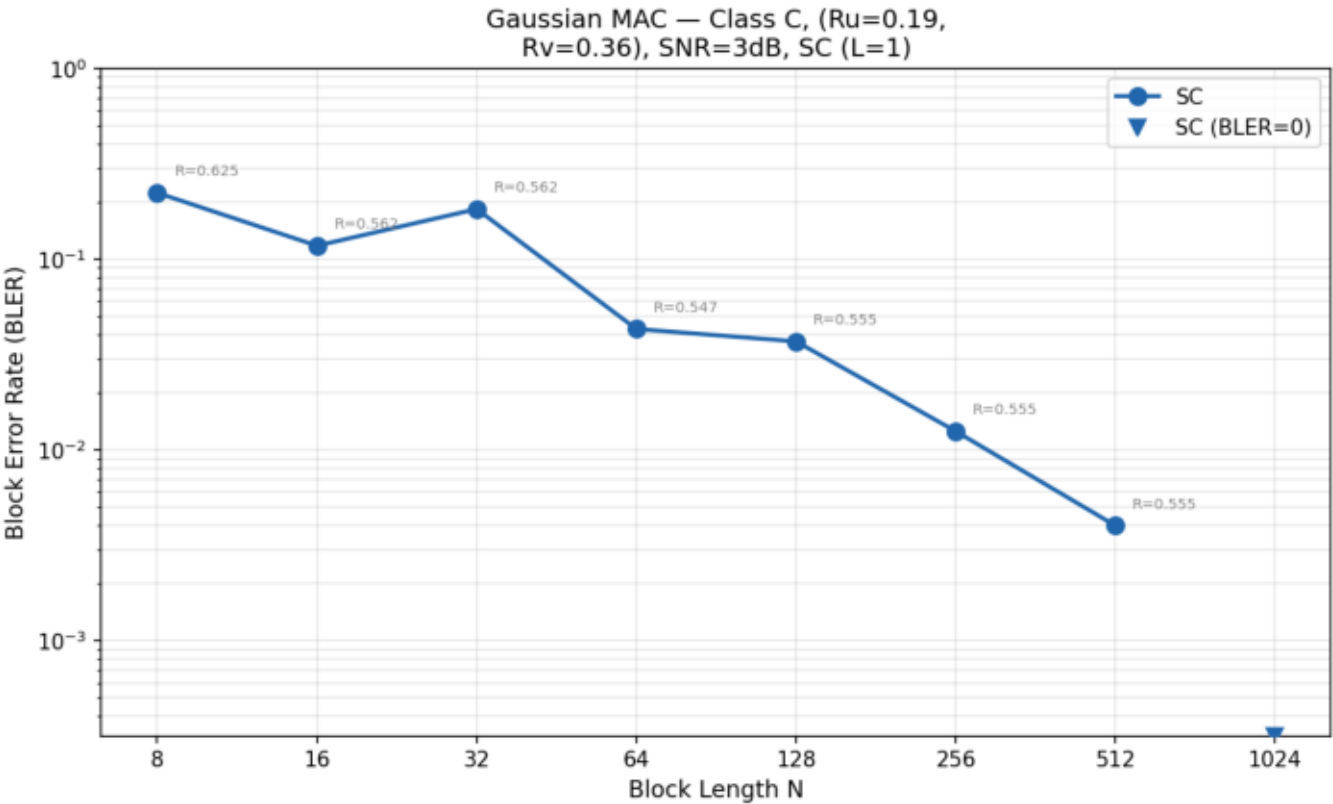
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	3	0.375	0.375	0.750	1.50e-01	5000
16	6	6	0.375	0.375	0.750	1.80e-01	5000
32	12	12	0.375	0.375	0.750	2.10e-01	5000
64	25	25	0.391	0.391	0.781	1.53e-01	5000
128	50	50	0.391	0.391	0.781	1.40e-01	10000
256	99	99	0.387	0.387	0.773	9.13e-02	10000
512	199	199	0.389	0.389	0.777	5.57e-02	10000
1024	398	398	0.389	0.389	0.777	2.31e-02	10000

[ALMOST_GOOD]

Comment: 5000+ cw, solid · almost-good

Gaussian MAC — Class C, (Ru=0.19, Rv=0.36), SNR=3dB, SC (L=1)

Symmetric capacity: Rx=Ry ≤ 0.5548 bits/use | Sum capacity: 1.1095 bits/use | Path: 0^N 1^N



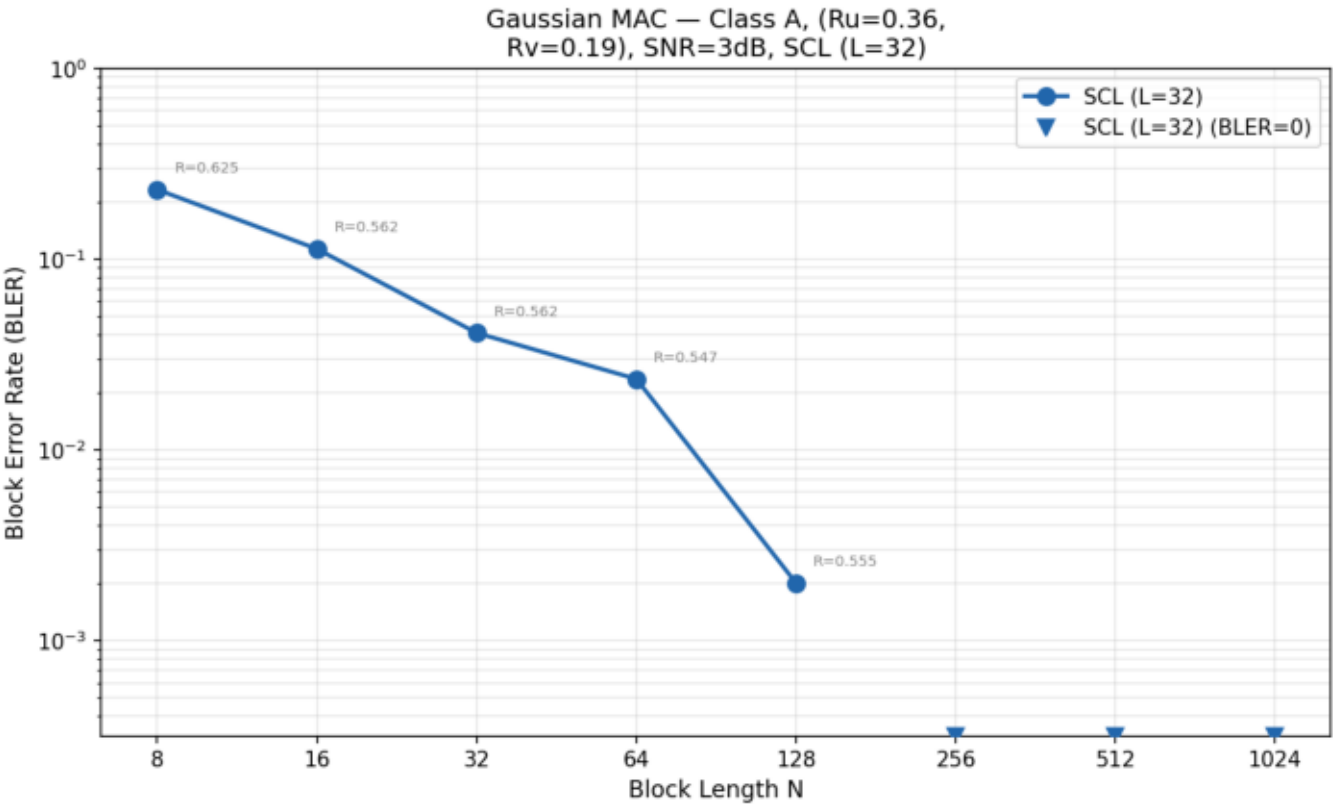
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	3	0.250	0.375	0.625	2.23e-01	2000
16	3	6	0.188	0.375	0.562	1.17e-01	2000
32	6	12	0.188	0.375	0.562	1.83e-01	2000
64	12	23	0.188	0.359	0.547	4.30e-02	2000
128	25	46	0.195	0.359	0.555	3.70e-02	2000
256	50	92	0.195	0.359	0.555	1.25e-02	2000
512	100	184	0.195	0.359	0.555	4.00e-03	2000
1024	199	369	0.194	0.360	0.555	0	1000

[ALMOST_GOOD]

Comment: 1000+ cw, moderate · almost-good

Gaussian MAC — Class A, (Ru=0.36, Rv=0.19), SNR=3dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.5548 bits/use | Sum capacity: 1.1095 bits/use | Path: 0^(3N/8) 1^N 0^(5N/8)



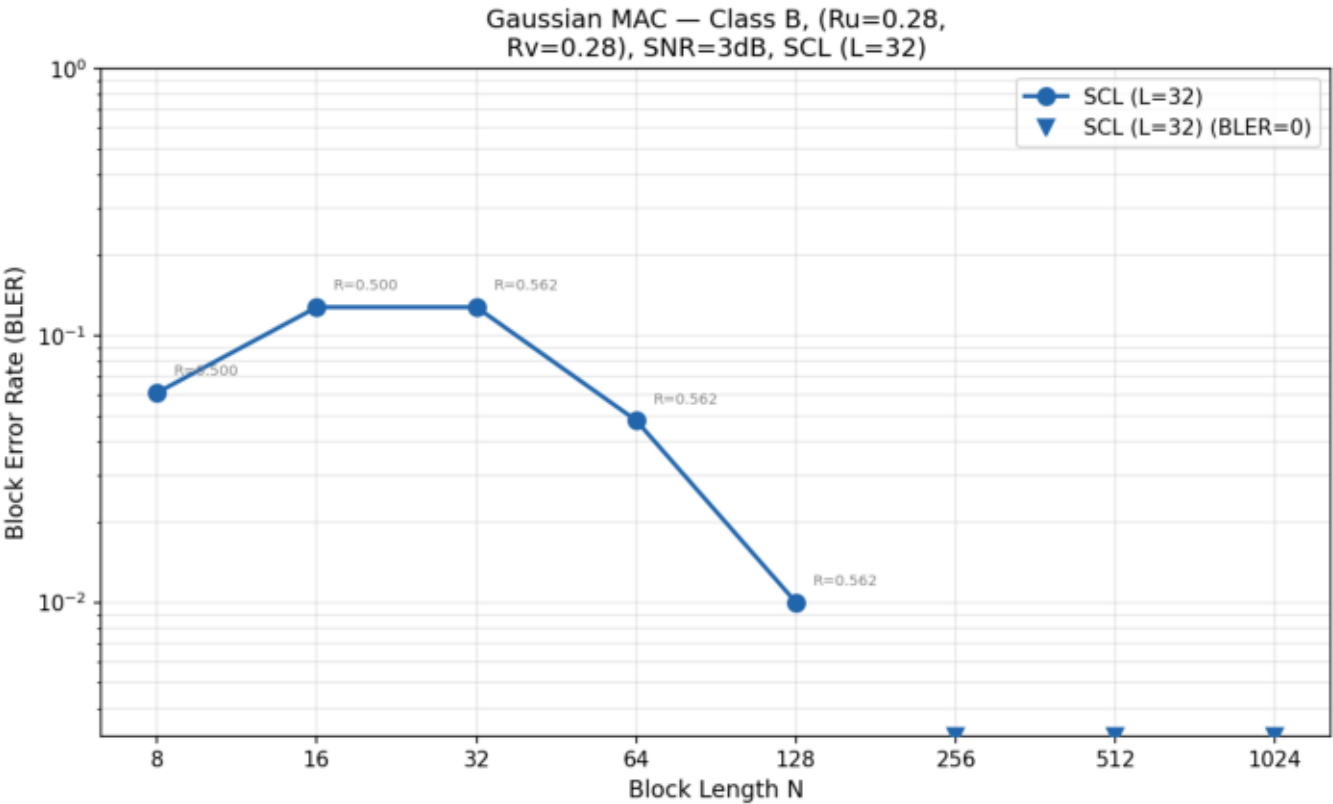
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	2	0.375	0.250	0.625	2.32e-01	2000
16	6	3	0.375	0.188	0.562	1.13e-01	2000
32	12	6	0.375	0.188	0.562	4.10e-02	2000
64	23	12	0.359	0.188	0.547	2.35e-02	2000
128	46	25	0.359	0.195	0.555	2.00e-03	2000
256	92	50	0.359	0.195	0.555	0	1000
512	184	100	0.359	0.195	0.555	0	1000
1024	369	199	0.360	0.194	0.555	0	500

[ALMOST_GOOD]

Comment: only 500 cw, low confidence · almost-good

Gaussian MAC — Class B, (Ru=0.28, Rv=0.28), SNR=3dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.5548 bits/use | Sum capacity: 1.1095 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



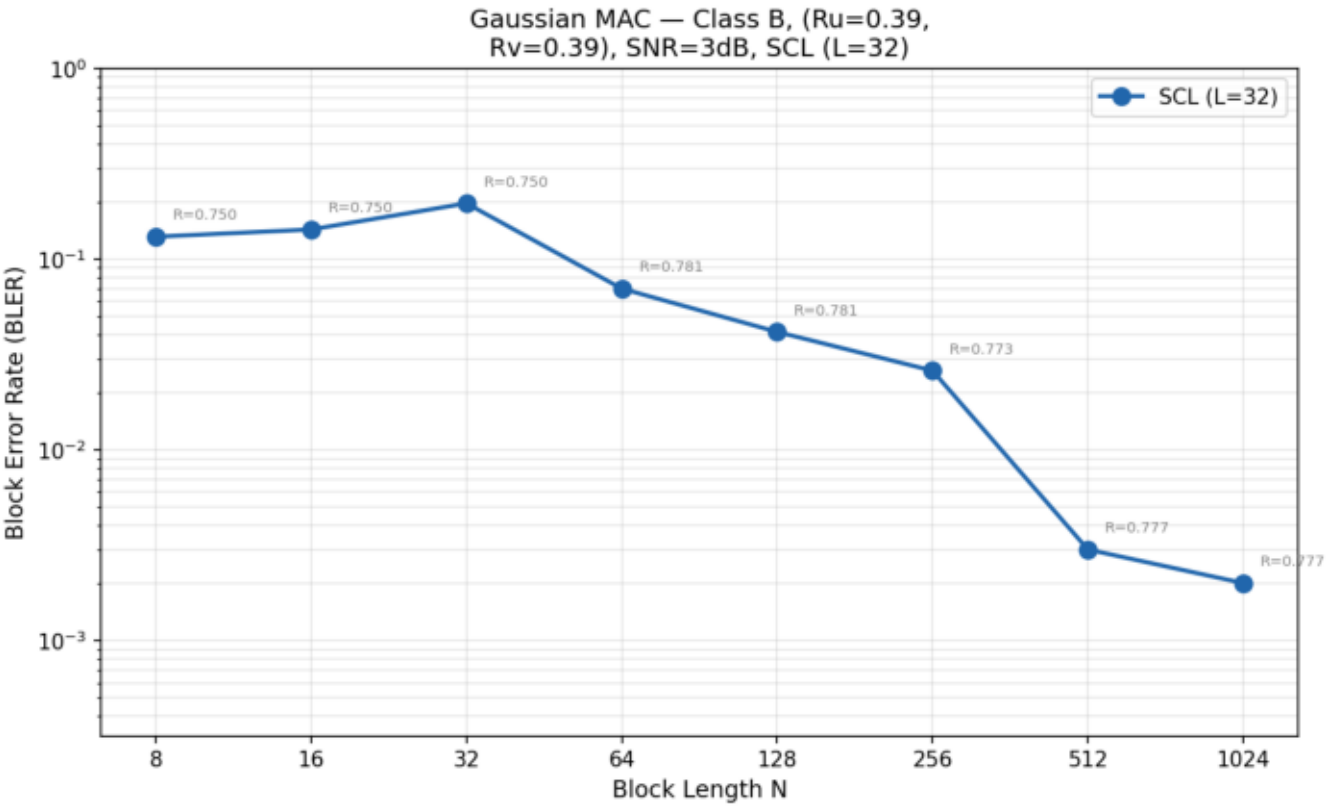
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	2	0.250	0.250	0.500	6.10e-02	2000
16	4	4	0.250	0.250	0.500	1.28e-01	2000
32	9	9	0.281	0.281	0.562	1.28e-01	2000
64	18	18	0.281	0.281	0.562	4.80e-02	2000
128	36	36	0.281	0.281	0.562	1.00e-02	2000
256	71	71	0.277	0.277	0.555	0	1000
512	142	142	0.277	0.277	0.555	0	1000
1024	284	284	0.277	0.277	0.555	0	500

[ALMOST_GOOD]

Comment: only 500 cw, low confidence · almost-good

Gaussian MAC — Class B, (Ru=0.39, Rv=0.39), SNR=3dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.5548 bits/use | Sum capacity: 1.1095 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



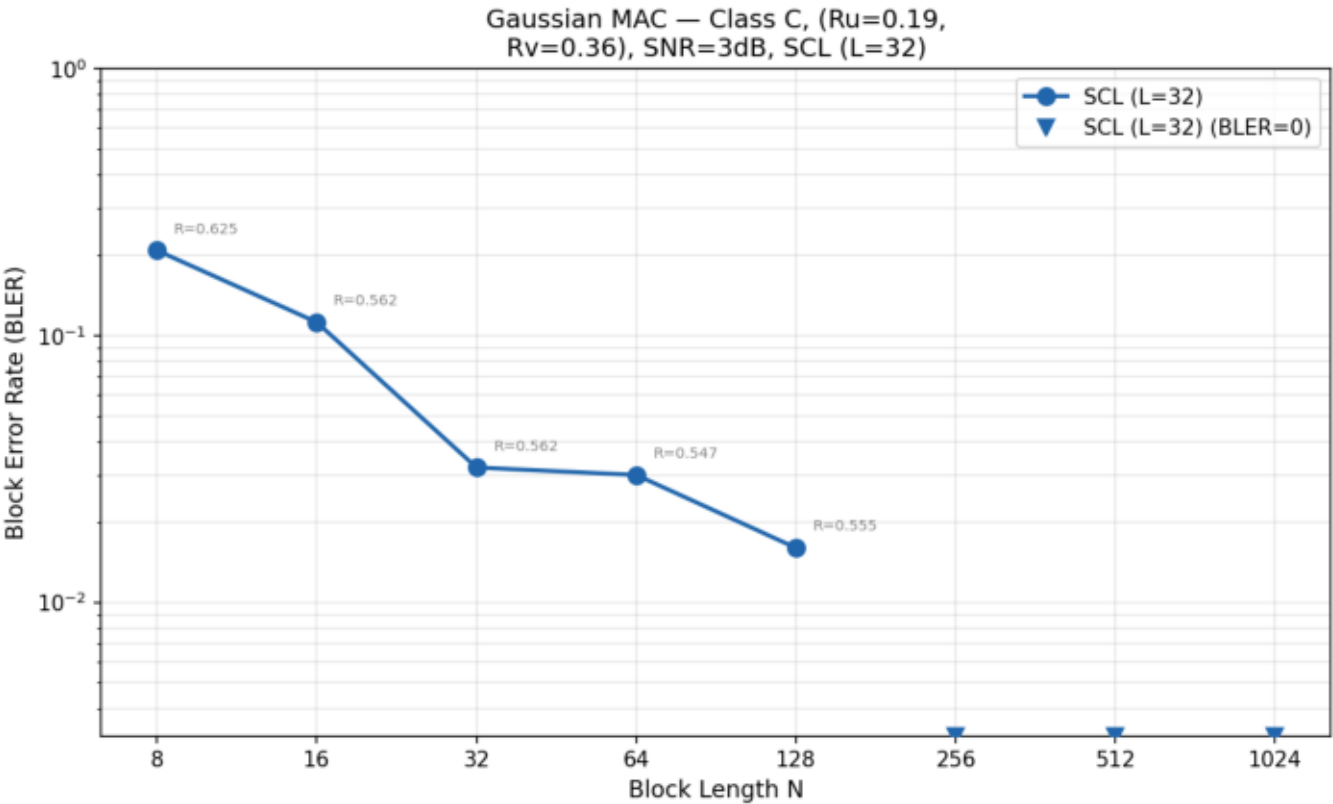
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	3	0.375	0.375	0.750	1.31e-01	2000
16	6	6	0.375	0.375	0.750	1.43e-01	2000
32	12	12	0.375	0.375	0.750	1.96e-01	2000
64	25	25	0.391	0.391	0.781	7.00e-02	2000
128	50	50	0.391	0.391	0.781	4.15e-02	2000
256	99	99	0.387	0.387	0.773	2.60e-02	1000
512	199	199	0.389	0.389	0.777	3.00e-03	1000
1024	398	398	0.389	0.389	0.777	2.00e-03	500

[ALMOST_GOOD]

Comment: only 500 cw, low confidence · almost-good

Gaussian MAC – Class C, (Ru=0.19, Rv=0.36), SNR=3dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.5548 bits/use | Sum capacity: 1.1095 bits/use | Path: 0^N 1^N



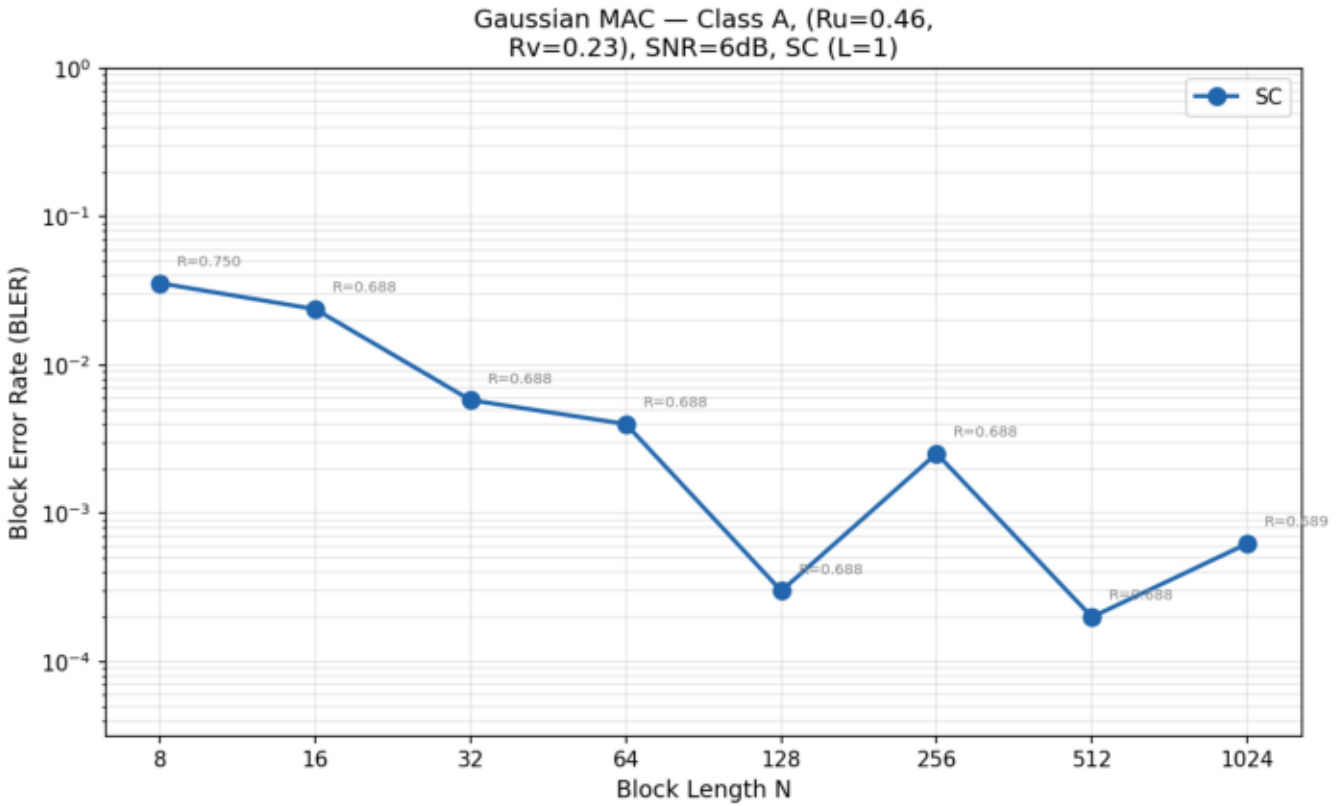
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	3	0.250	0.375	0.625	2.08e-01	500
16	3	6	0.188	0.375	0.562	1.12e-01	500
32	6	12	0.188	0.375	0.562	3.20e-02	500
64	12	23	0.188	0.359	0.547	3.00e-02	500
128	25	46	0.195	0.359	0.555	1.60e-02	500
256	50	92	0.195	0.359	0.555	0	200
512	100	184	0.195	0.359	0.555	0	200
1024	199	369	0.194	0.360	0.555	0	100

[GOOD]

Comment: only 100 cw, low confidence · classified GOOD

Gaussian MAC — Class A, (Ru=0.46, Rv=0.23), SNR=6dB, SC (L=1)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^(3N/8) 1^N 0^(5N/8)



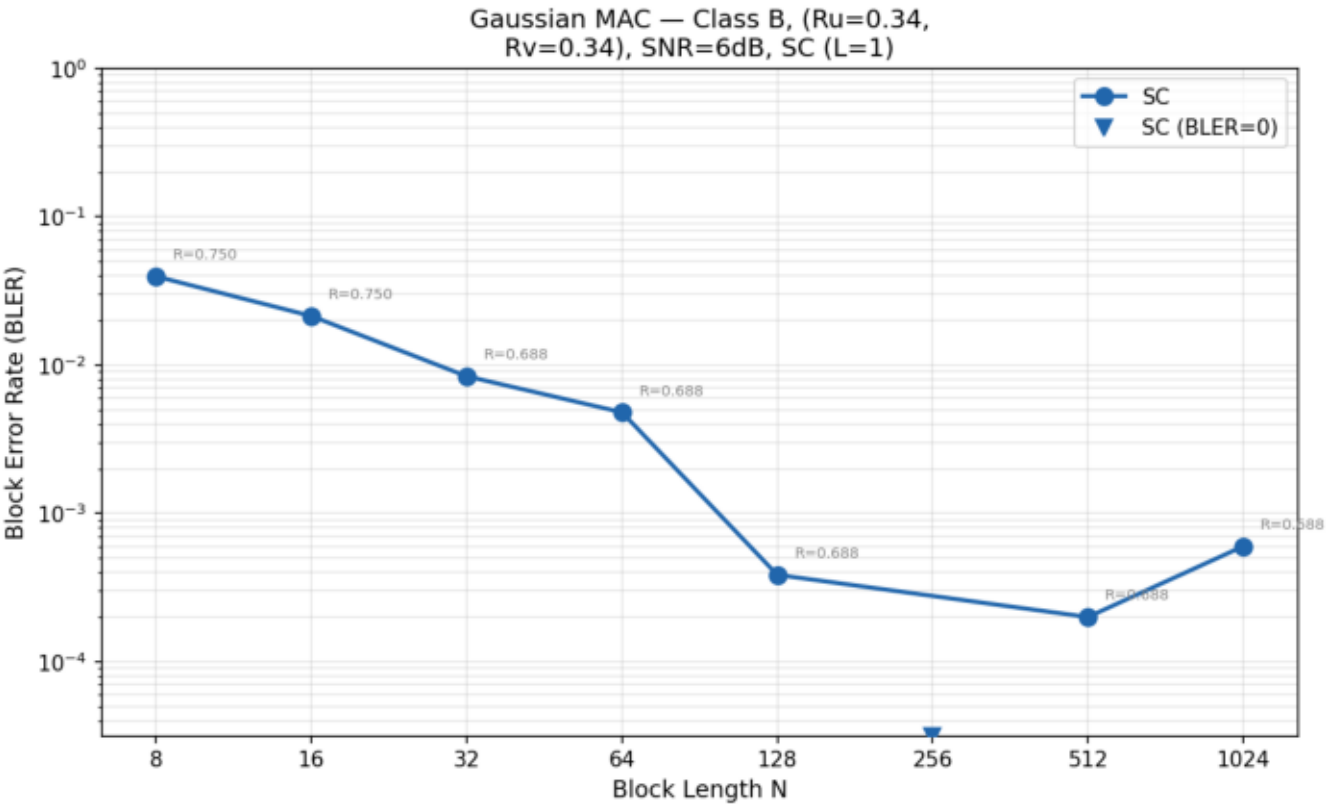
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	4	2	0.500	0.250	0.750	3.56e-02	5000
16	7	4	0.438	0.250	0.688	2.38e-02	5000
32	15	7	0.469	0.219	0.688	5.80e-03	5000
64	29	15	0.453	0.234	0.688	4.00e-03	5000
128	58	30	0.453	0.234	0.688	3.00e-04	20000
256	117	59	0.457	0.231	0.688	2.53e-03	15000
512	233	119	0.455	0.232	0.688	2.00e-04	10000
1024	467	238	0.456	0.232	0.689	6.25e-04	8000

[WEIRD]

Comment: 5000+ cw, solid · flagged WEIRD · non-monotonic (likely small-N rate quantization)

Gaussian MAC — Class B, (Ru=0.34, Rv=0.34), SNR=6dB, SC (L=1)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



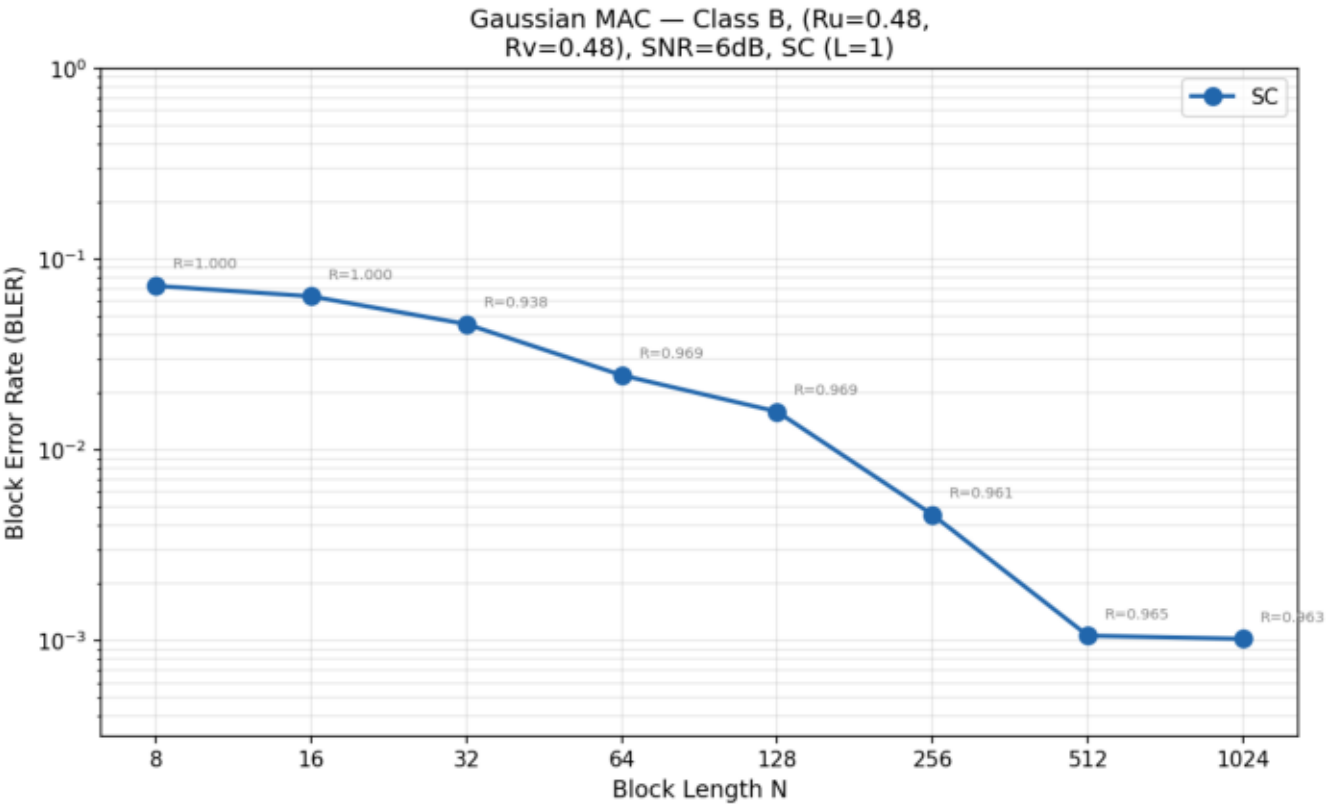
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	3	0.375	0.375	0.750	3.96e-02	5000
16	6	6	0.375	0.375	0.750	2.14e-02	5000
32	11	11	0.344	0.344	0.688	8.40e-03	5000
64	22	22	0.344	0.344	0.688	4.80e-03	5000
128	44	44	0.344	0.344	0.688	3.85e-04	13000
256	88	88	0.344	0.344	0.688	0	13000
512	176	176	0.344	0.344	0.688	2.00e-04	10000
1024	352	352	0.344	0.344	0.688	6.00e-04	10000

[WEIRD]

Comment: 5000+ cw, solid · flagged WEIRD · non-monotonic (likely small-N rate quantization)

Gaussian MAC — Class B, (Ru=0.48, Rv=0.48), SNR=6dB, SC (L=1)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



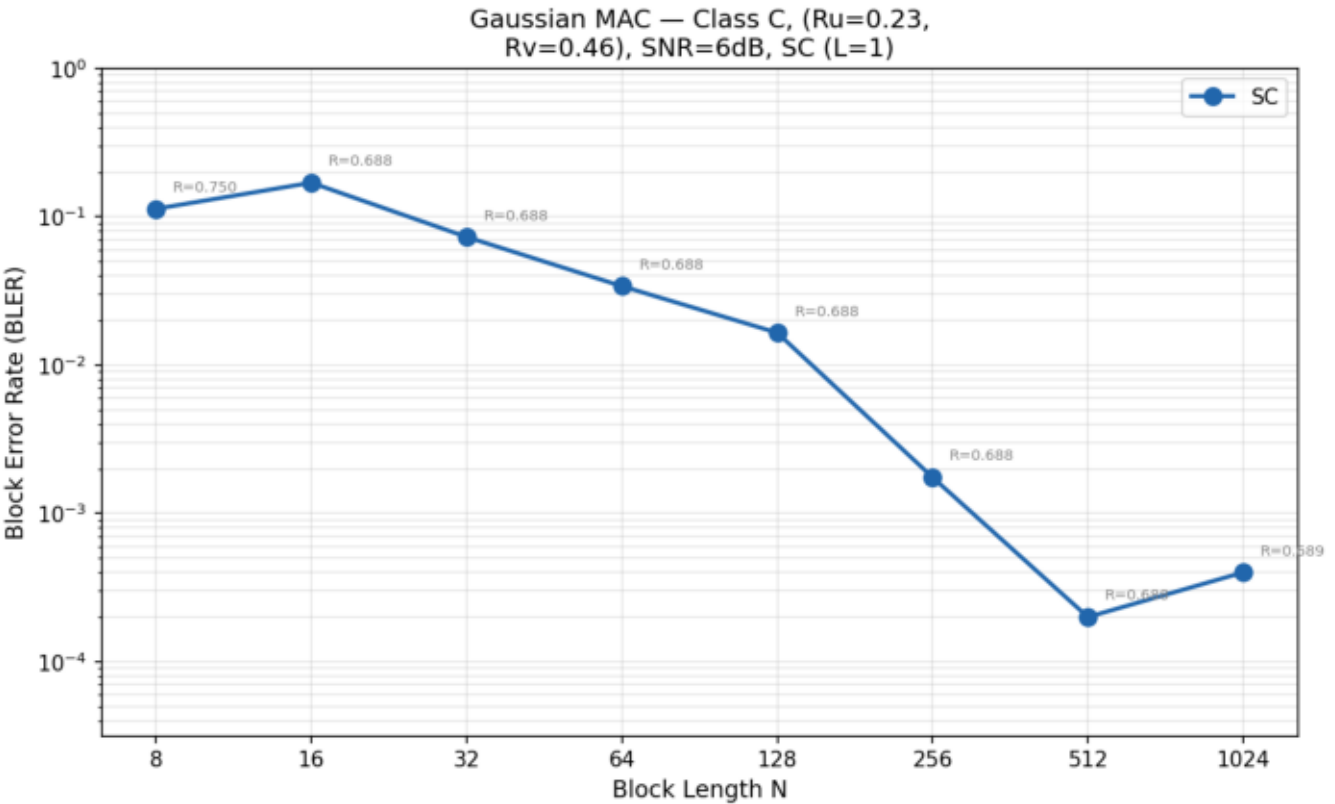
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	4	4	0.500	0.500	1	7.24e-02	5000
16	8	8	0.500	0.500	1	6.38e-02	5000
32	15	15	0.469	0.469	0.938	4.56e-02	5000
64	31	31	0.484	0.484	0.969	2.46e-02	5000
128	62	62	0.484	0.484	0.969	1.58e-02	20000
256	123	123	0.480	0.480	0.961	4.57e-03	30000
512	247	247	0.482	0.482	0.965	1.06e-03	50000
1024	493	493	0.481	0.481	0.963	1.02e-03	50000

[GOOD]

Comment: 5000+ cw, solid · classified GOOD

Gaussian MAC — Class C, (Ru=0.23, Rv=0.46), SNR=6dB, SC (L=1)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^N 1^N



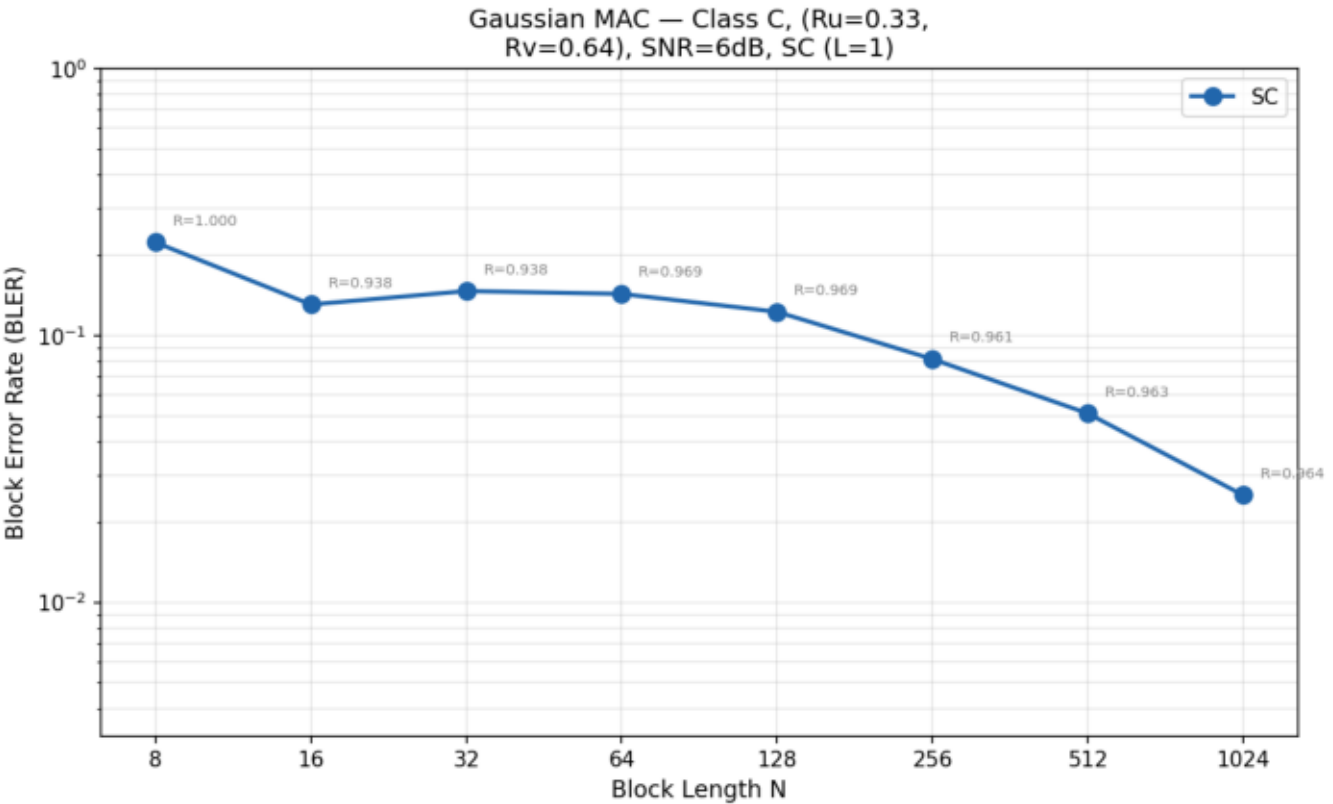
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	4	0.250	0.500	0.750	1.13e-01	2000
16	4	7	0.250	0.438	0.688	1.70e-01	2000
32	7	15	0.219	0.469	0.688	7.30e-02	2000
64	15	29	0.234	0.453	0.688	3.40e-02	2000
128	30	58	0.234	0.453	0.688	1.65e-02	2000
256	59	117	0.231	0.457	0.688	1.75e-03	20000
512	119	233	0.232	0.455	0.688	2.00e-04	20000
1024	238	467	0.232	0.456	0.689	4.00e-04	10000

[WEIRD]

Comment: 2000+ cw, moderate · flagged WEIRD · non-monotonic (likely small-N rate quantization)

Gaussian MAC — Class C, (Ru=0.33, Rv=0.64), SNR=6dB, SC (L=1)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^N 1^N



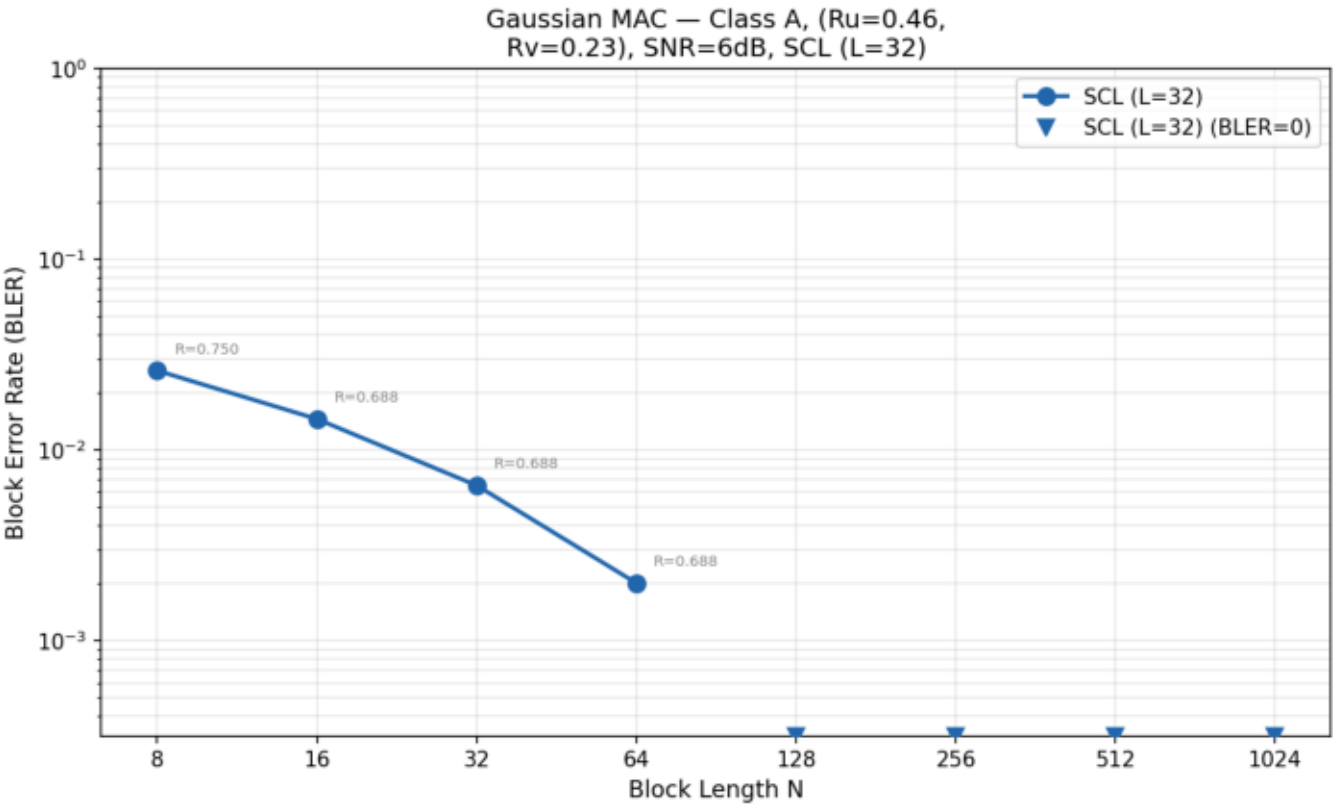
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	5	0.375	0.625	1	2.23e-01	2000
16	5	10	0.312	0.625	0.938	1.31e-01	2000
32	10	20	0.312	0.625	0.938	1.46e-01	2000
64	21	41	0.328	0.641	0.969	1.43e-01	2000
128	42	82	0.328	0.641	0.969	1.22e-01	2000
256	83	163	0.324	0.637	0.961	8.15e-02	2000
512	166	327	0.324	0.639	0.963	5.10e-02	2000
1024	333	654	0.325	0.639	0.964	2.52e-02	5000

[ALMOST_GOOD]

Comment: 2000+ cw, moderate · almost-good

Gaussian MAC — Class A, (Ru=0.46, Rv=0.23), SNR=6dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^(3N/8) 1^N 0^(5N/8)



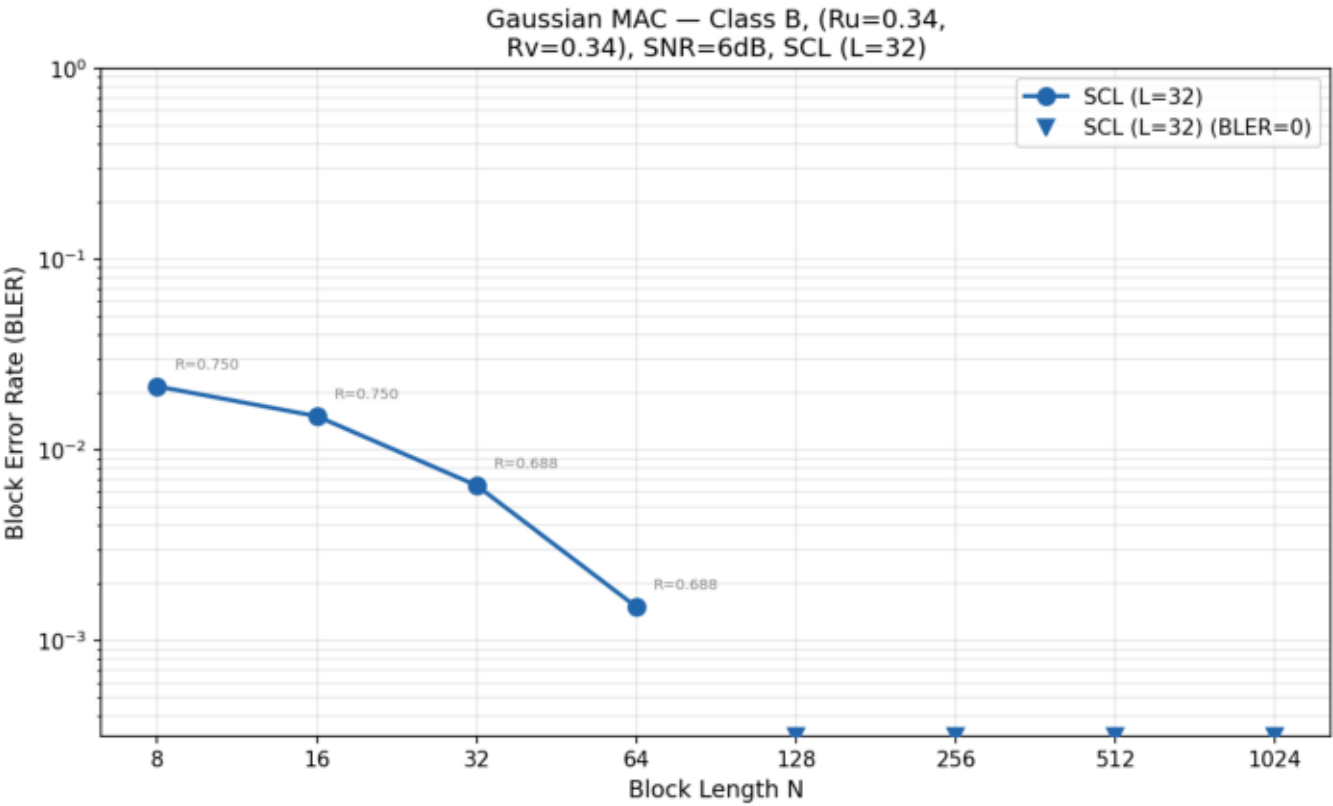
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	4	2	0.500	0.250	0.750	2.60e-02	2000
16	7	4	0.438	0.250	0.688	1.45e-02	2000
32	15	7	0.469	0.219	0.688	6.50e-03	2000
64	29	15	0.453	0.234	0.688	2.00e-03	2000
128	58	30	0.453	0.234	0.688	0	2000
256	117	59	0.457	0.231	0.688	0	1000
512	233	119	0.455	0.232	0.688	0	1000
1024	467	238	0.456	0.232	0.689	0	500

[ALMOST_GOOD]

Comment: only 500 cw, low confidence · almost-good

Gaussian MAC — Class B, (Ru=0.34, Rv=0.34), SNR=6dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



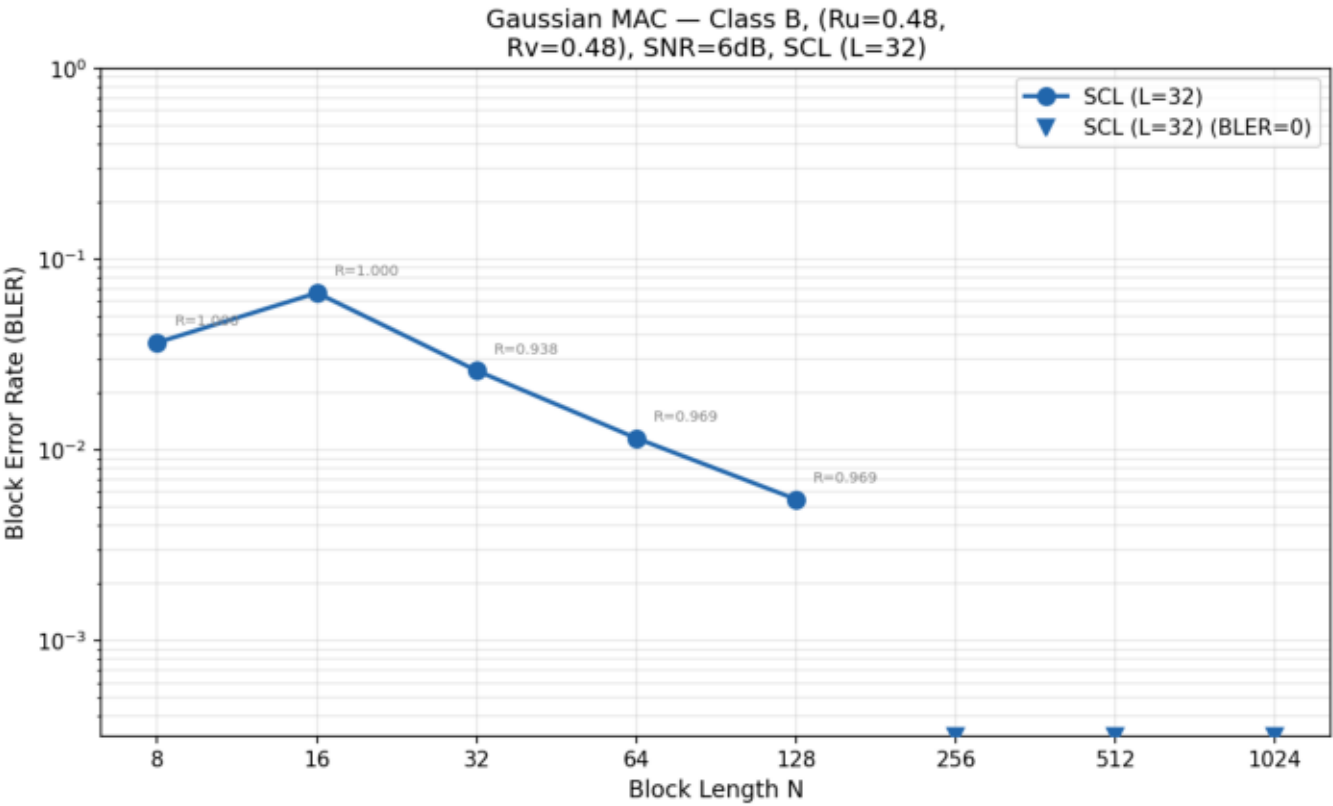
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	3	0.375	0.375	0.750	2.15e-02	2000
16	6	6	0.375	0.375	0.750	1.50e-02	2000
32	11	11	0.344	0.344	0.688	6.50e-03	2000
64	22	22	0.344	0.344	0.688	1.50e-03	2000
128	44	44	0.344	0.344	0.688	0	2000
256	88	88	0.344	0.344	0.688	0	1000
512	176	176	0.344	0.344	0.688	0	1000
1024	352	352	0.344	0.344	0.688	0	500

[ALMOST_GOOD]

Comment: only 500 cw, low confidence · almost-good

Gaussian MAC — Class B, (Ru=0.48, Rv=0.48), SNR=6dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



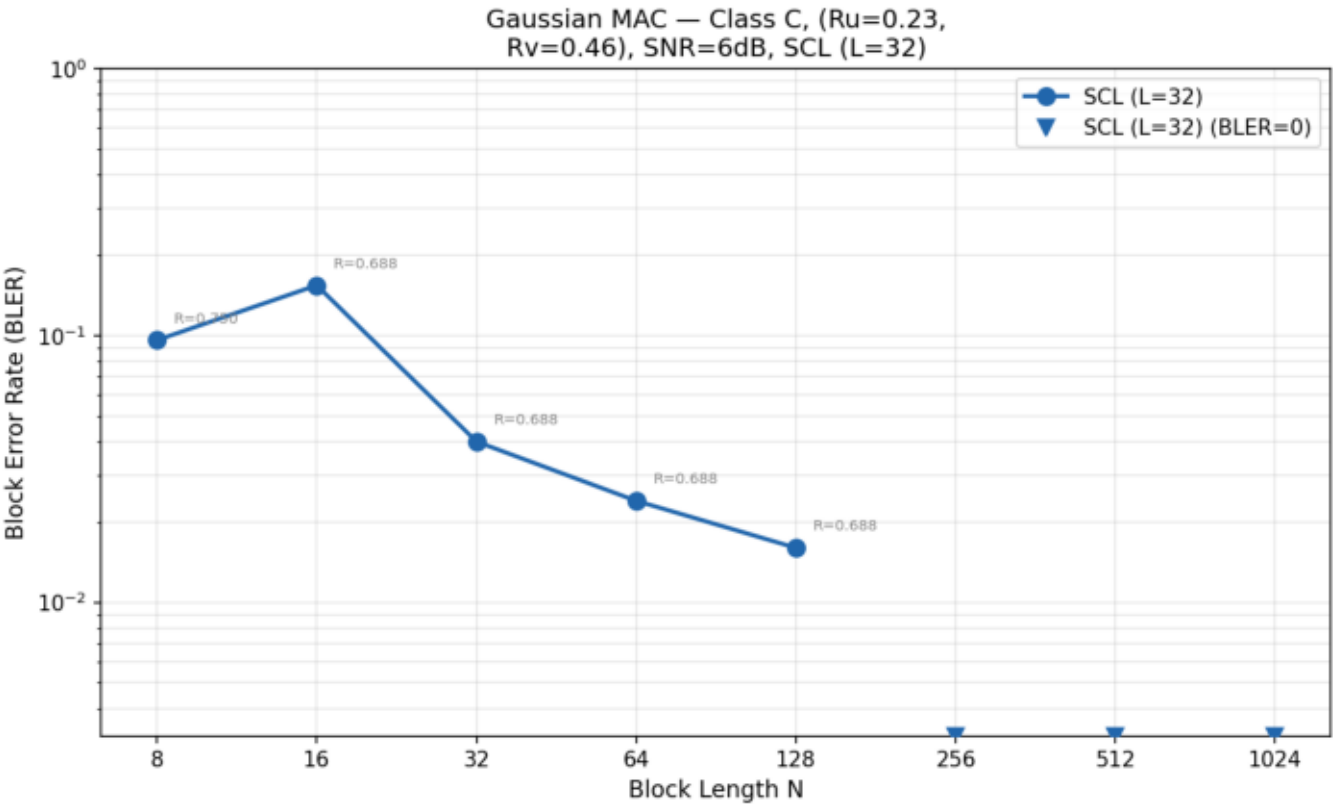
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	4	4	0.500	0.500	1	3.65e-02	2000
16	8	8	0.500	0.500	1	6.65e-02	2000
32	15	15	0.469	0.469	0.938	2.60e-02	2000
64	31	31	0.484	0.484	0.969	1.15e-02	2000
128	62	62	0.484	0.484	0.969	5.50e-03	2000
256	123	123	0.480	0.480	0.961	0	1000
512	247	247	0.482	0.482	0.965	0	1000
1024	493	493	0.481	0.481	0.963	0	500

[ALMOST_GOOD]

Comment: only 500 cw, low confidence · almost-good

Gaussian MAC — Class C, (Ru=0.23, Rv=0.46), SNR=6dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^N 1^N



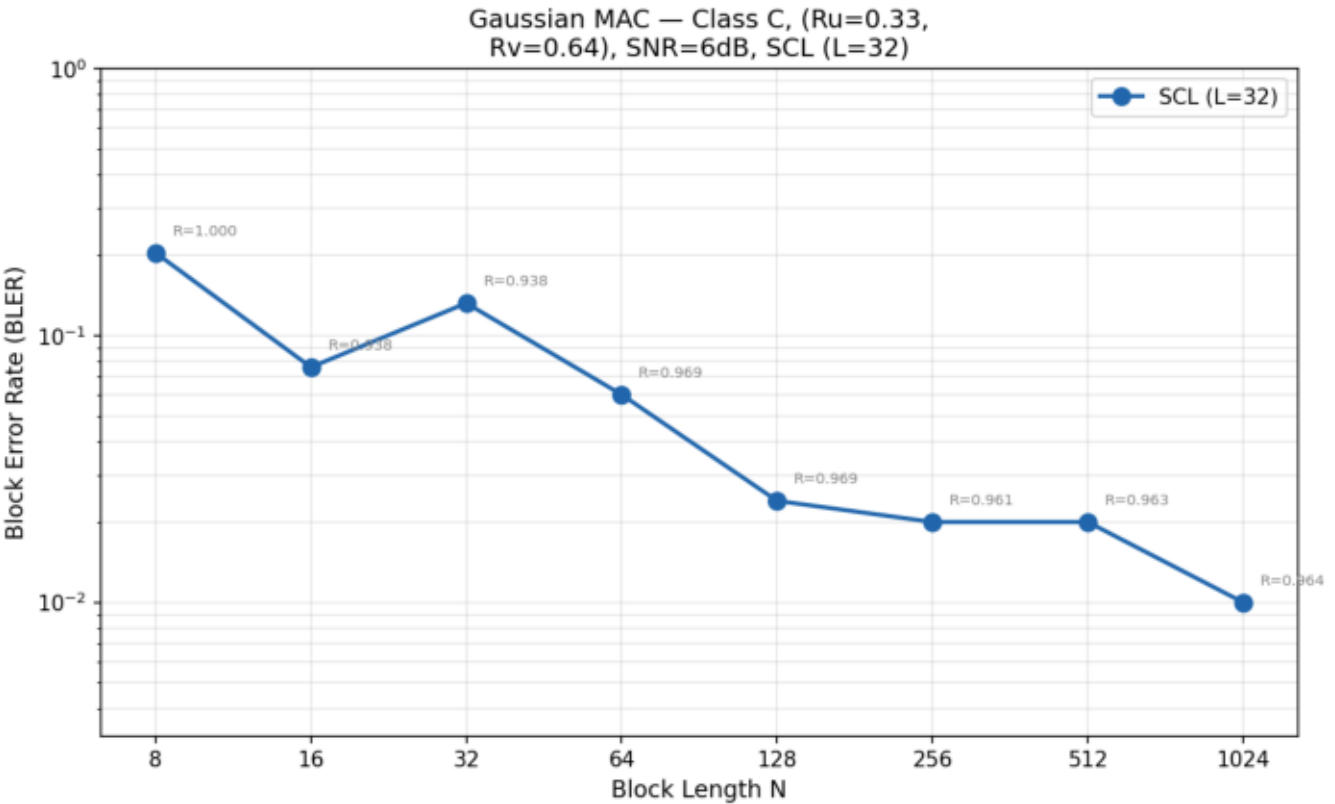
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	4	0.250	0.500	0.750	9.60e-02	500
16	4	7	0.250	0.438	0.688	1.54e-01	500
32	7	15	0.219	0.469	0.688	4.00e-02	500
64	15	29	0.234	0.453	0.688	2.40e-02	500
128	30	58	0.234	0.453	0.688	1.60e-02	500
256	59	117	0.231	0.457	0.688	0	200
512	119	233	0.232	0.455	0.688	0	200
1024	238	467	0.232	0.456	0.689	0	100

[ALMOST_GOOD]

Comment: only 100 cw, low confidence · almost-good

Gaussian MAC — Class C, (Ru=0.33, Rv=0.64), SNR=6dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^N 1^N



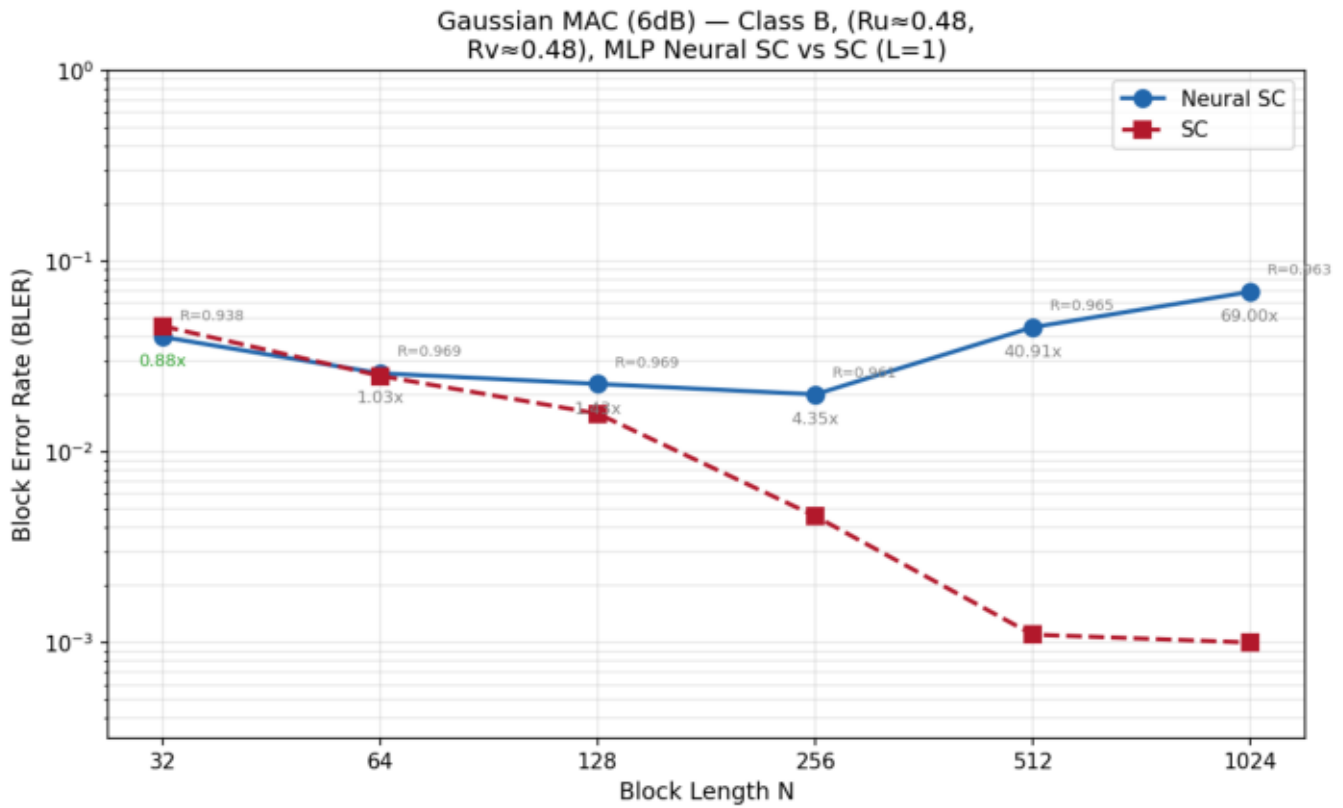
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	3	5	0.375	0.625	1	2.04e-01	500
16	5	10	0.312	0.625	0.938	7.60e-02	500
32	10	20	0.312	0.625	0.938	1.32e-01	500
64	21	41	0.328	0.641	0.969	6.00e-02	500
128	42	82	0.328	0.641	0.969	2.40e-02	500
256	83	163	0.324	0.637	0.961	2.00e-02	200
512	166	327	0.324	0.639	0.963	2.00e-02	200
1024	333	654	0.325	0.639	0.964	1.00e-02	100

[ALMOST_GOOD]

Comment: only 100 cw, low confidence · almost-good

Gaussian MAC (6dB) — Class B, ($R_u \approx 0.48$, $R_v \approx 0.48$), MLP Neural SC vs SC (L=1)

Capacity: Sym cap/user=0.6882 | z_encoder: MLP(1→32→16) | Path: $U^{(N/2)} V^N U^{(N/2)}$



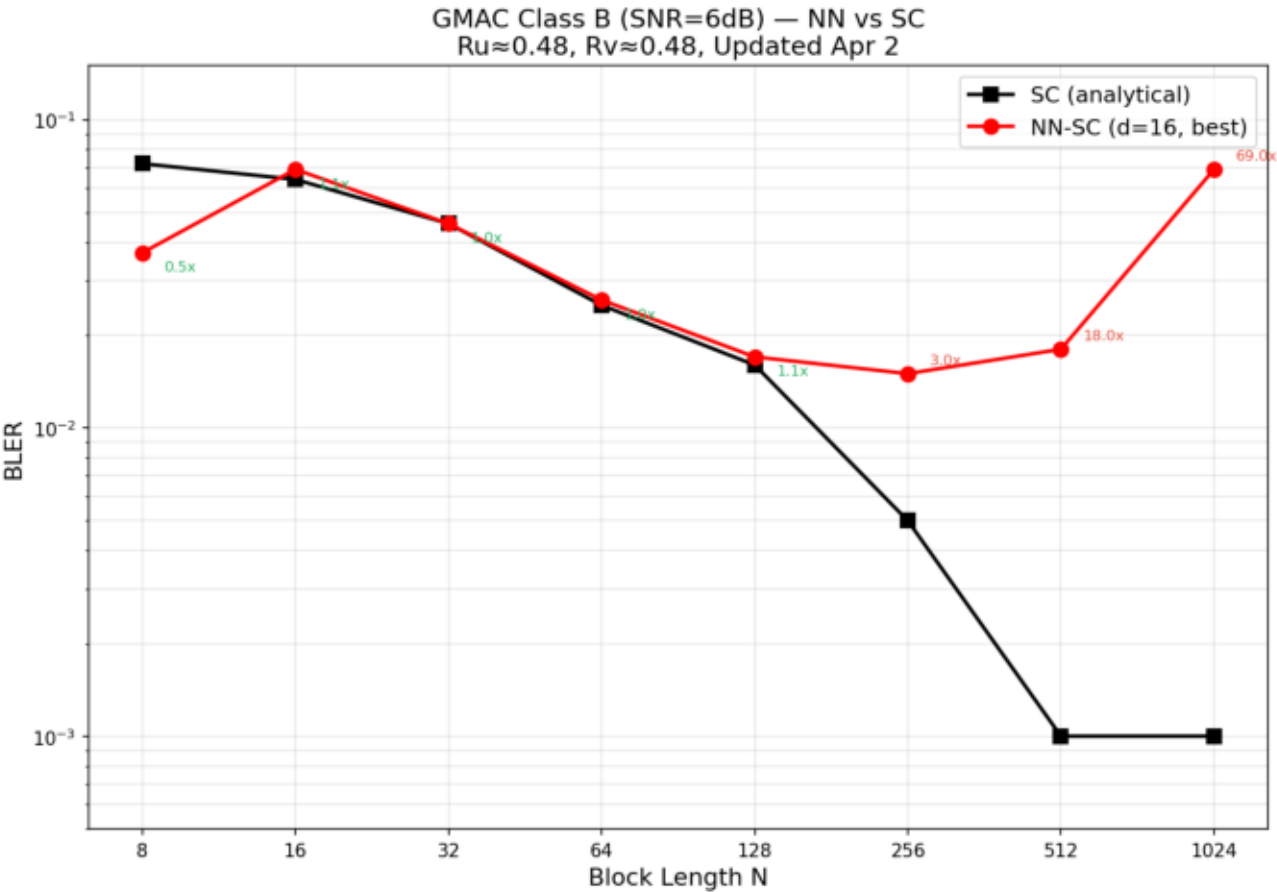
N	ku	kv	Ru	Rv	NN BLER	SC BLER	NN/SC	Codewords
32	15	15	0.469	0.469	4.00e-02	4.56e-02	0.880	5000
64	31	31	0.484	0.484	2.58e-02	2.50e-02	1.030	50000
128	62	62	0.484	0.484	2.27e-02	1.58e-02	1.430	50000
256	123	123	0.480	0.480	2.00e-02	4.60e-03	4.350	50000
512	247	247	0.482	0.482	4.50e-02	1.10e-03	40.910	50000
1024	493	493	0.481	0.481	6.90e-02	1.00e-03	69	50000

[WEIRD]

Comment: 5000+ cw, solid · flagged WEIRD · NN close at small N, diverges large N (0.88–69.00x)

Gaussian MAC (6dB) — Class B, (Ru≈0.48, Rv≈0.48), Neural SC vs SC (L=1)

Capacity: I(Z;X)=0.4645 | I(Z;Y|X)=0.9119 | I(Z;X,Y)=1.3764 | Sym cap/user=0.6882 | Path: U^(N/2) V^N U^(N/2)



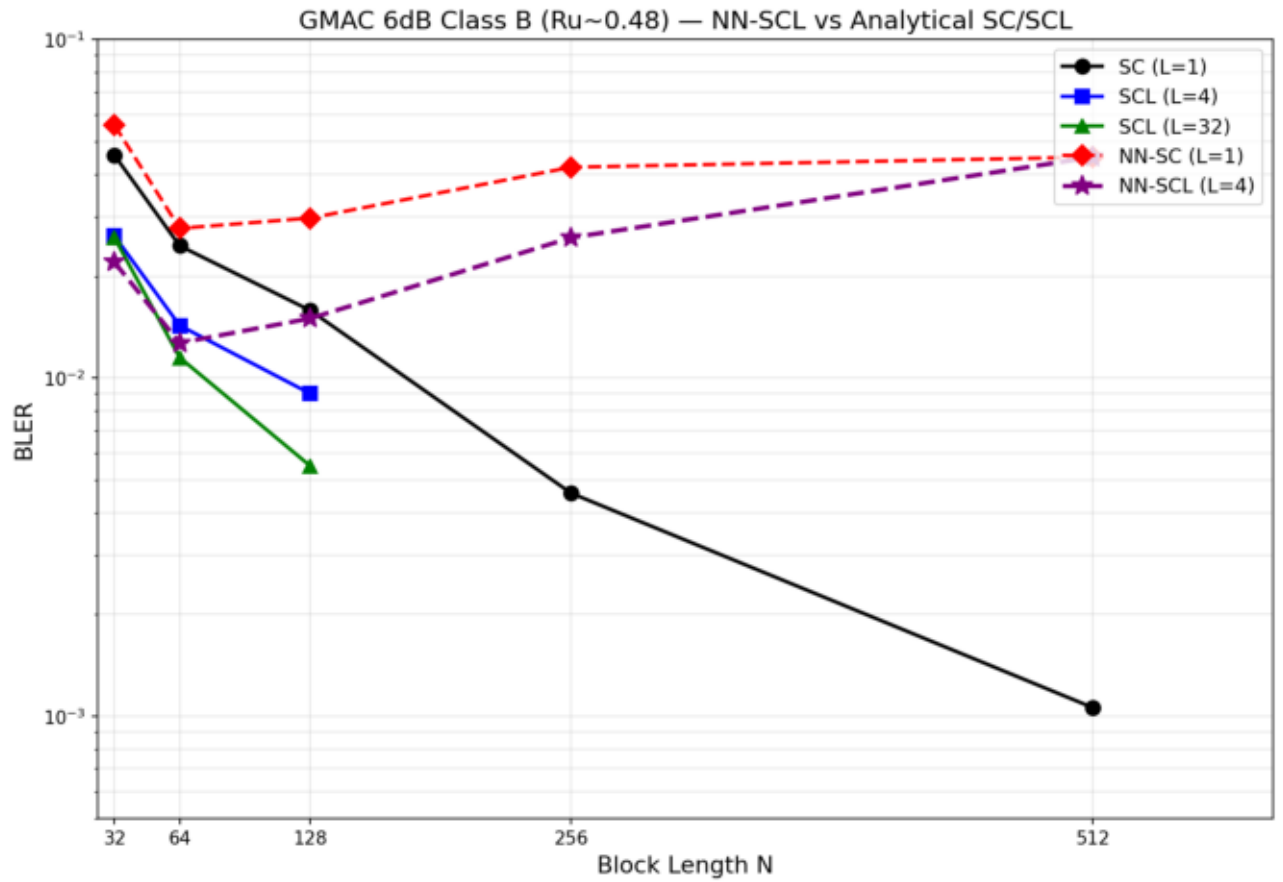
N	ku	kv	Ru	Rv	NN BLER	SC BLER	NN/SC	Codewords
8	4	4	0.500	0.500	3.70e-02	7.20e-02	0.510	5000
16	8	8	0.500	0.500	6.90e-02	6.40e-02	1.080	5000
32	15	15	0.469	0.469	4.60e-02	4.60e-02	1	3000
64	31	31	0.484	0.484	2.60e-02	2.50e-02	1.040	3000
128	62	62	0.484	0.484	1.70e-02	1.60e-02	1.060	3000
256	123	123	0.480	0.480	1.50e-02	5.00e-03	3	5000
512	247	247	0.482	0.482	1.80e-02	1.00e-03	18	1000
1024	493	493	0.481	0.481	6.90e-02	1.00e-03	69	200

[GOOD]

Comment: only 200 cw, low confidence · classified GOOD · NN close at small N, diverges large N (0.51–69.00x)

Gaussian MAC (6dB) — Class B, (Ru~0.48, Rv~0.48), NN-SCL vs SC/SCL Comparison

Path: $U^{\wedge}(N/2) V^{\wedge}N U^{\wedge}(N/2)$ | Model: SimpleMLP_Gmac (39K params) | Designs: MC-based NPZ



N	ku	kv	Ru	SC BLER	SC C	SCL (L=4) BLER	SCL (L=8) BLER	SCL (L=32) BLER	SCL (L=32) C	NN-SC BLER	NN-SC C	SCL (L=4) BLER	SCL (L=4) C	NN-SCL BLER	NN-SCL C
32	15	15	0.469	4.56e-02	5000	2.63e-02	3000	2.60e-02	2000	5.60e-02	3000	2.20e-02	3000	0.48	0.84
64	31	31	0.484	2.46e-02	5000	1.43e-02	3000	1.15e-02	2000	2.77e-02	3000	1.27e-02	3000	0.52	0.89
128	62	62	0.484	1.58e-02	20000	9.00e-03	3000	5.50e-03	2000	2.97e-02	3000	1.50e-02	1000	0.95	1.67
256	123	123	0.480	4.57e-03	30000					4.20e-02	500	2.60e-02	500	5.69	
512	247	247	0.482	1.06e-03	50000					4.50e-02	200	4.50e-02	200	42.45	

[UNCATEGORIZED]

Comment: 5000+ cw, solid